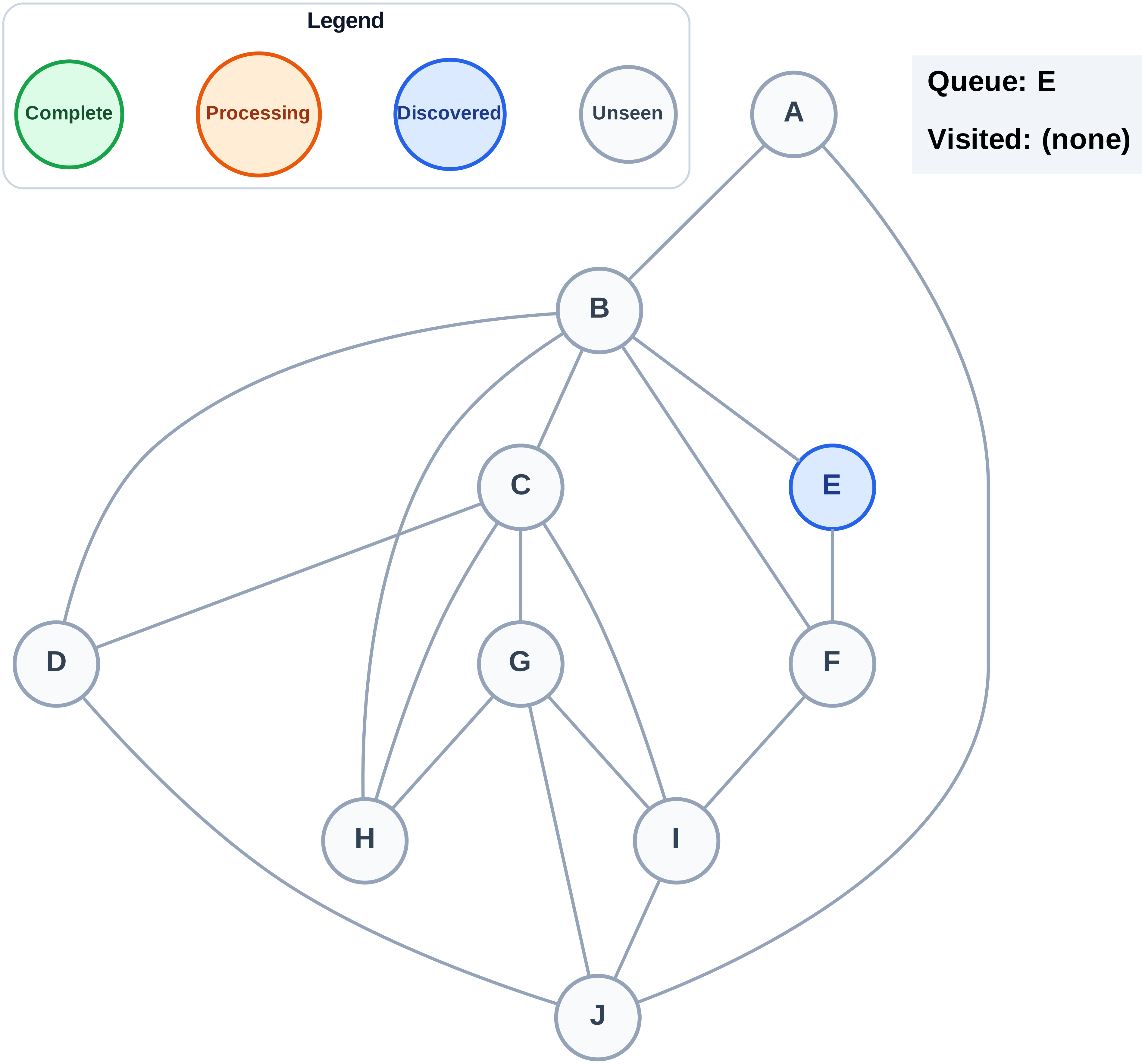
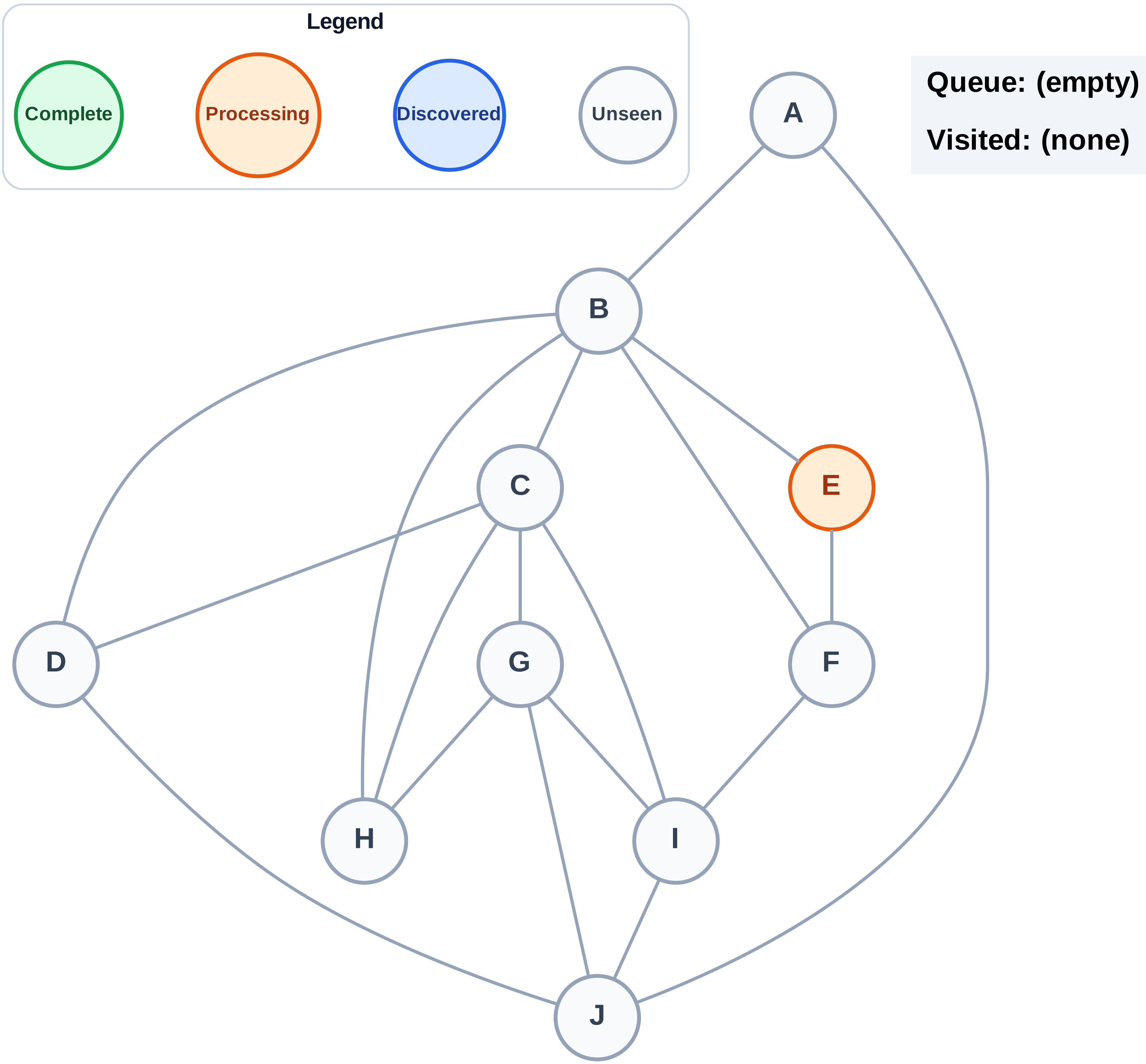
BFS Traversal

Step 1: Start at E



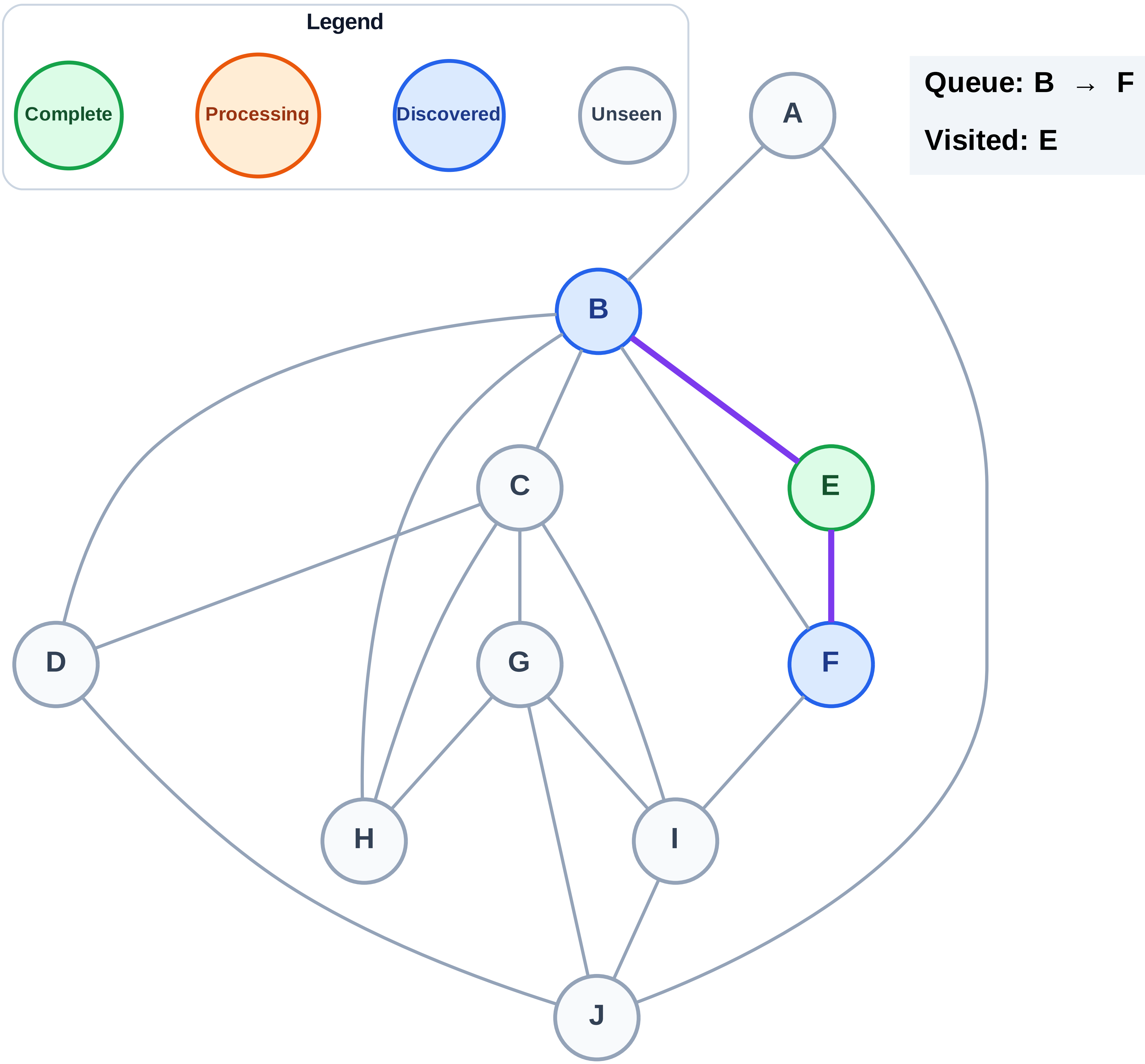
BFS Traversal

Step 2: Process node E



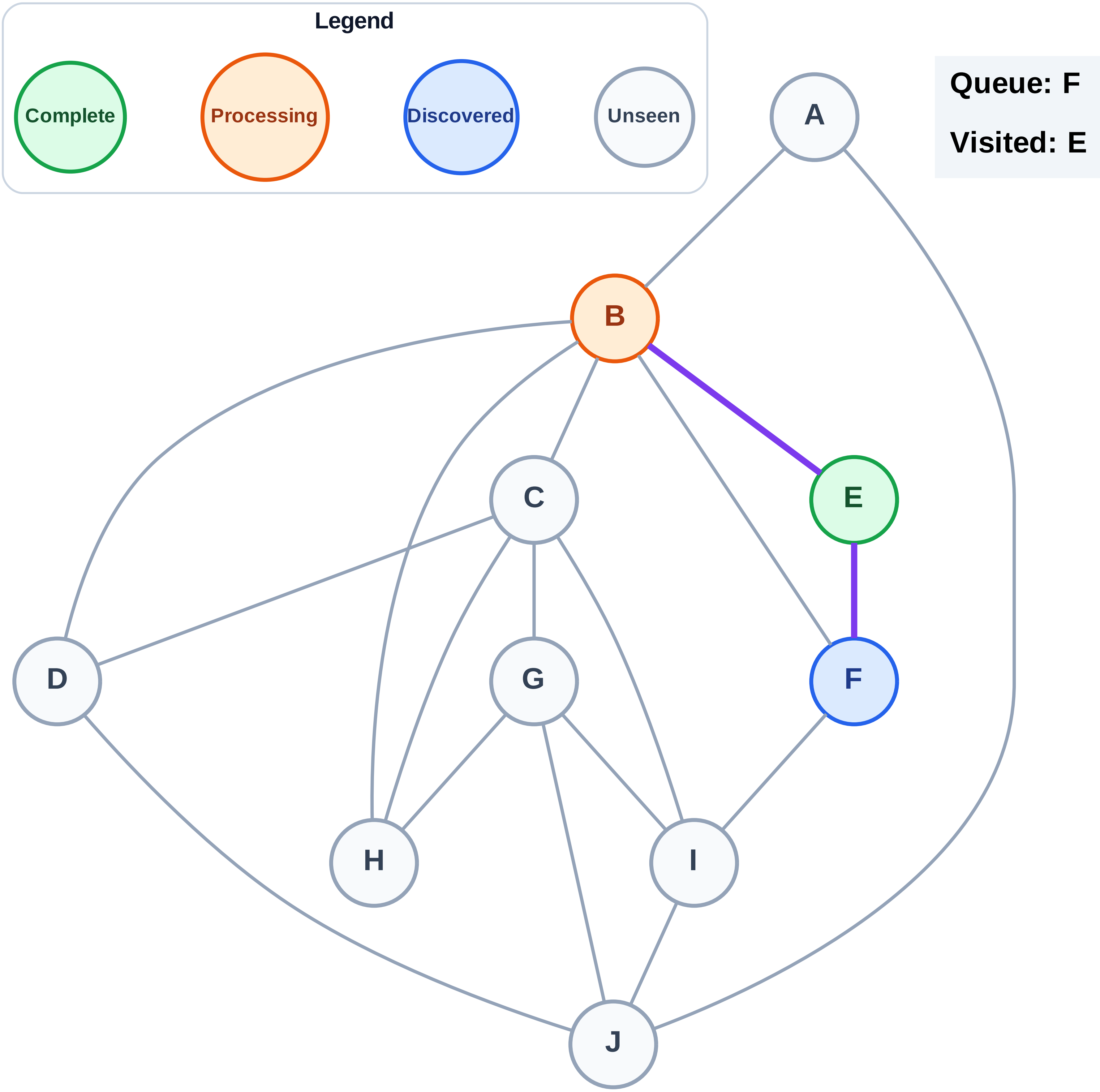
BFS Traversal

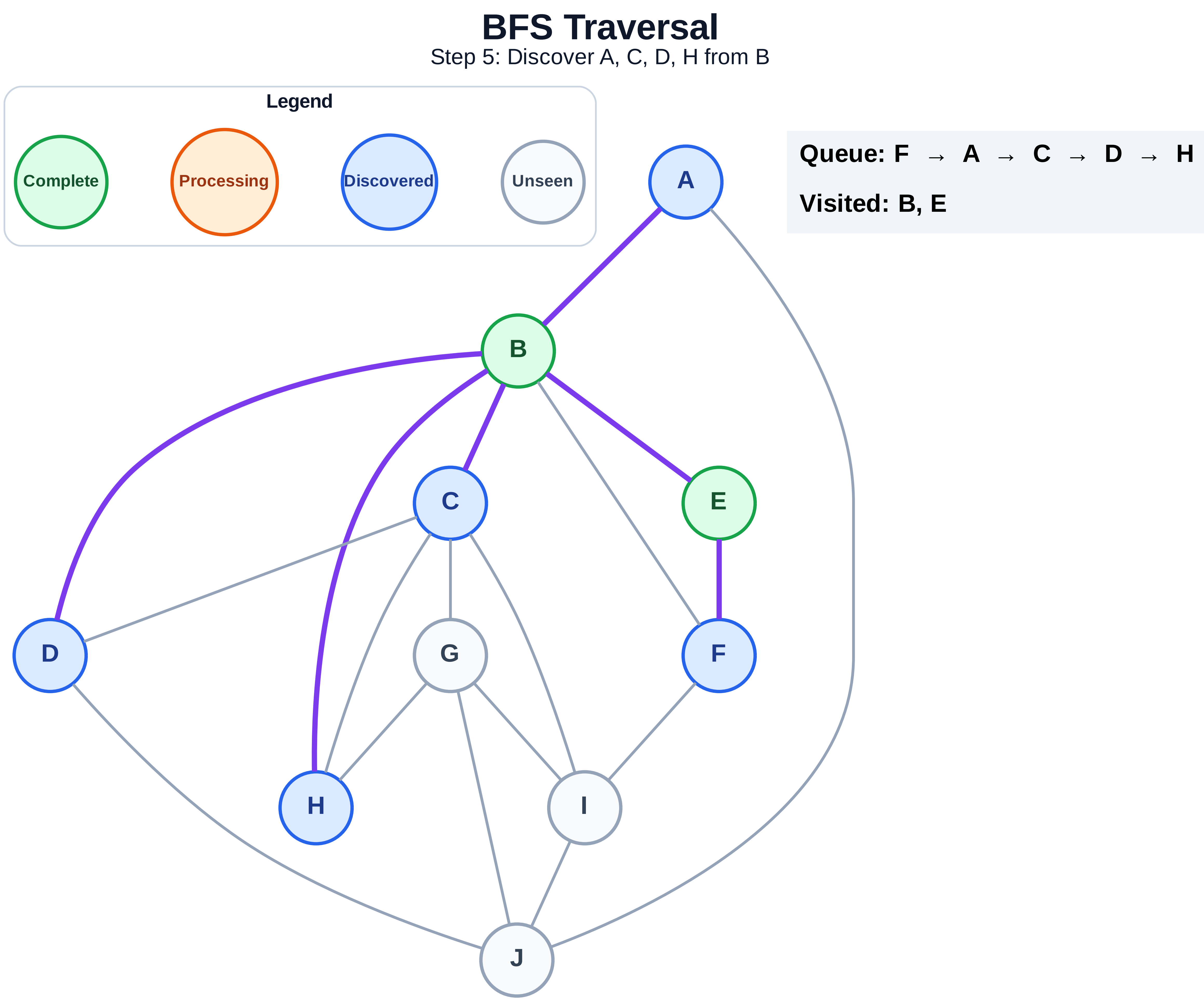
Step 3: Discover B, F from E



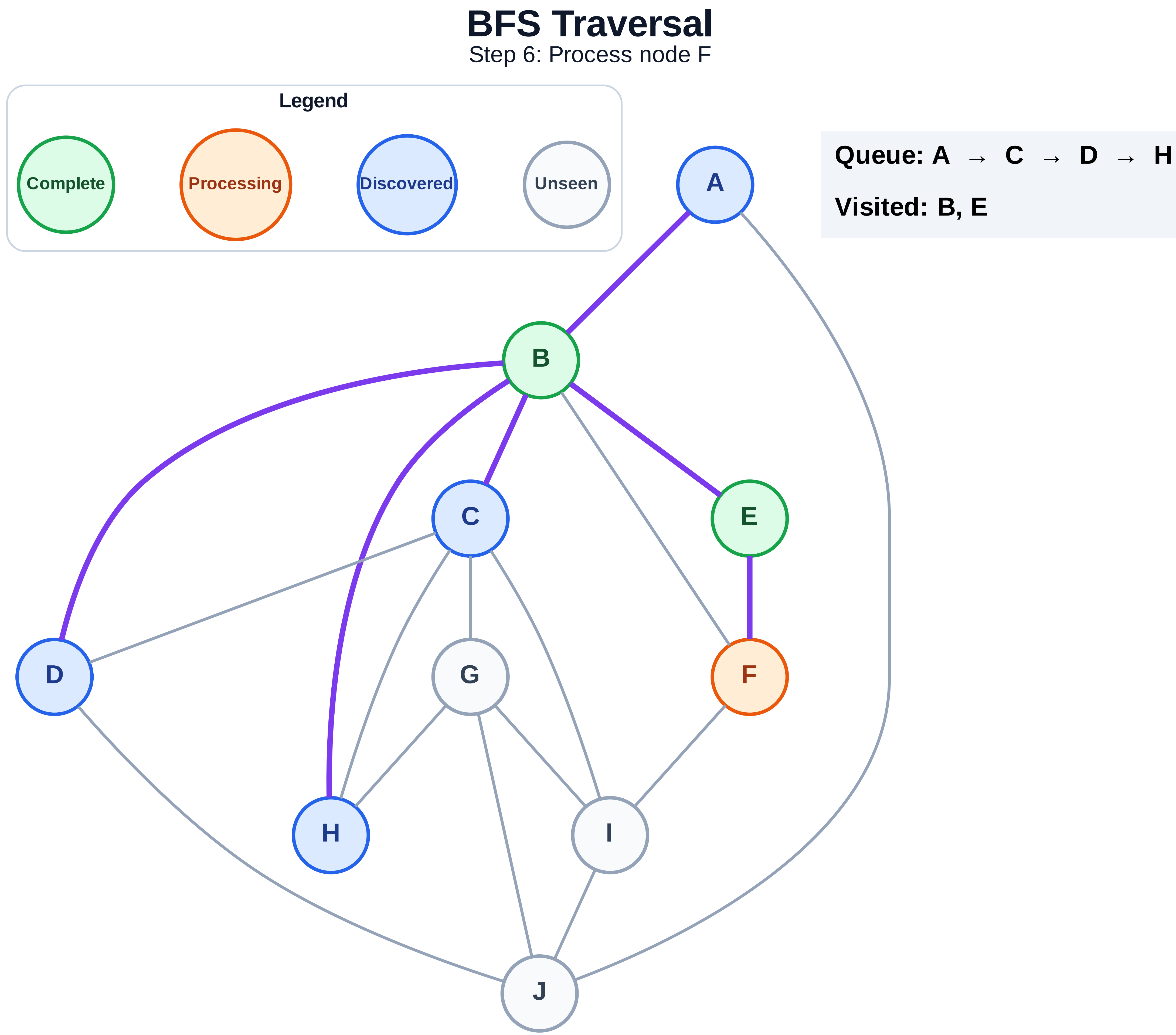
BFS Traversal

Step 4: Process node B



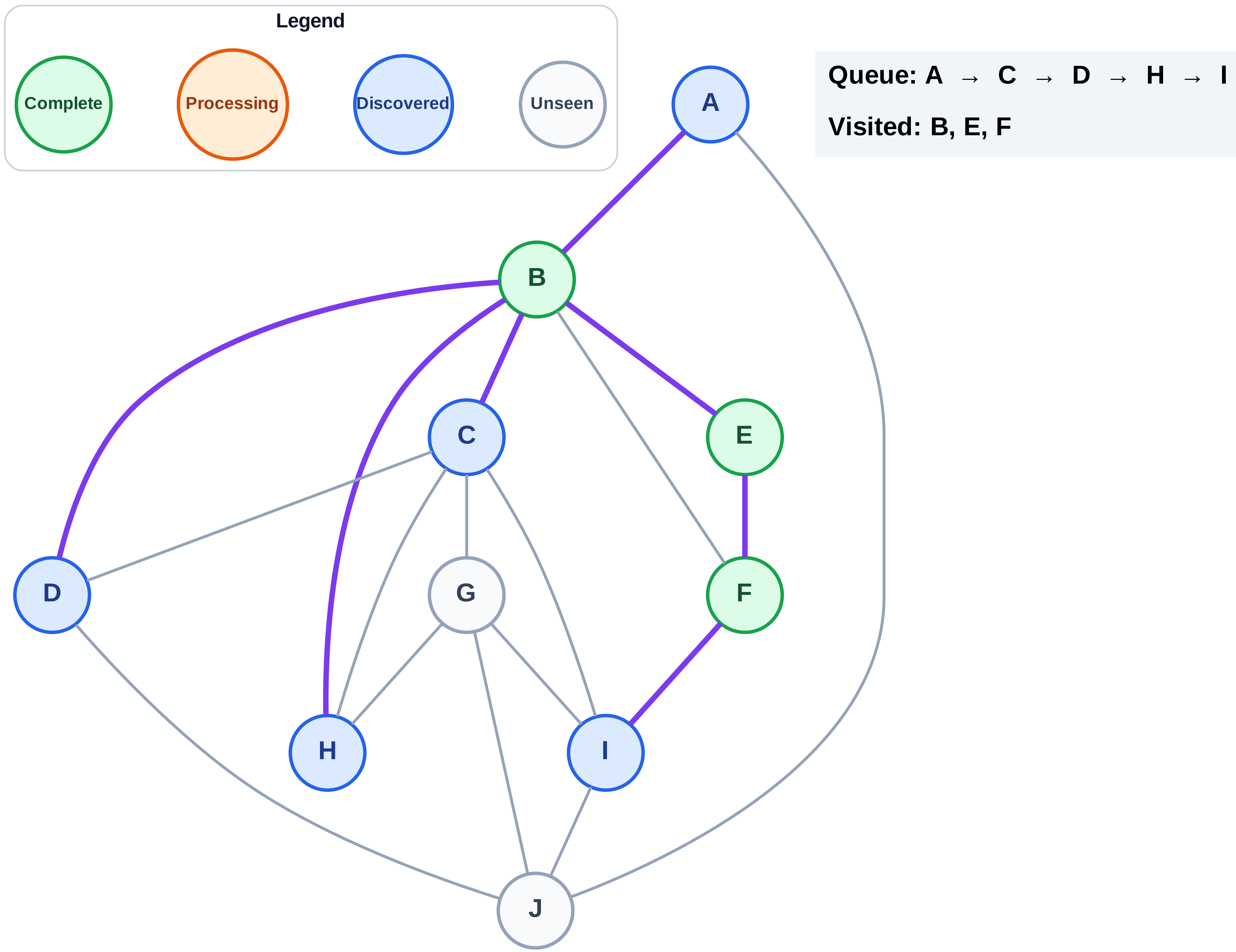


BFS Traversal
Step 5: Discover A, C, D, H from B
Legend
Complete
Processing
Discovered
Unseen
Queue: F → A → C → D → H
Visited: B, E
A
B
C
D
E
F
G
H
I
J


BFS Traversal
Step 6: Process node F
Legend
Complete
Processing
Discovered
Unseen
Queue: A → C → D → H
Visited: B, E
A
B
C
D
E
F
G
H
I
J

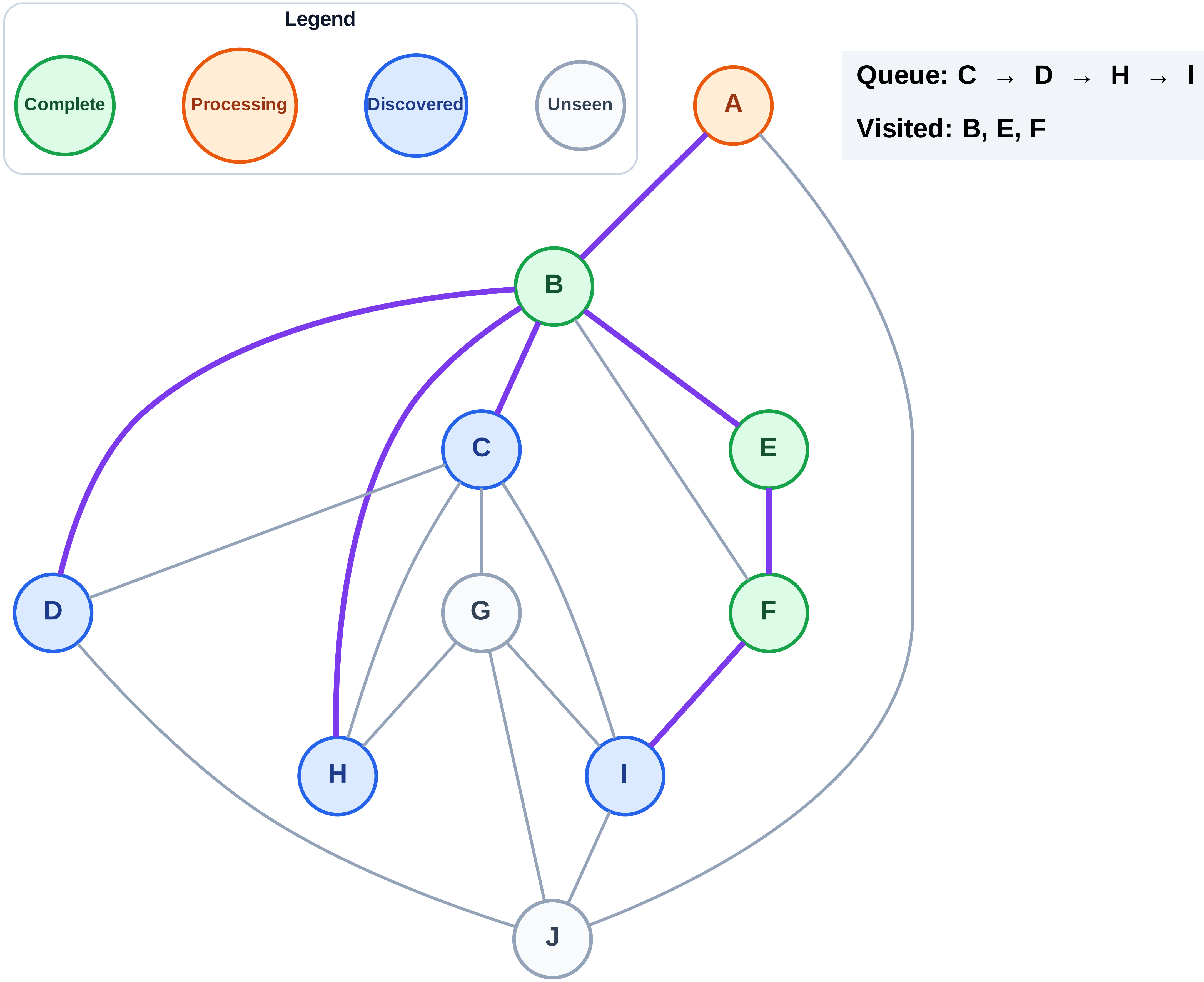
BFS Traversal

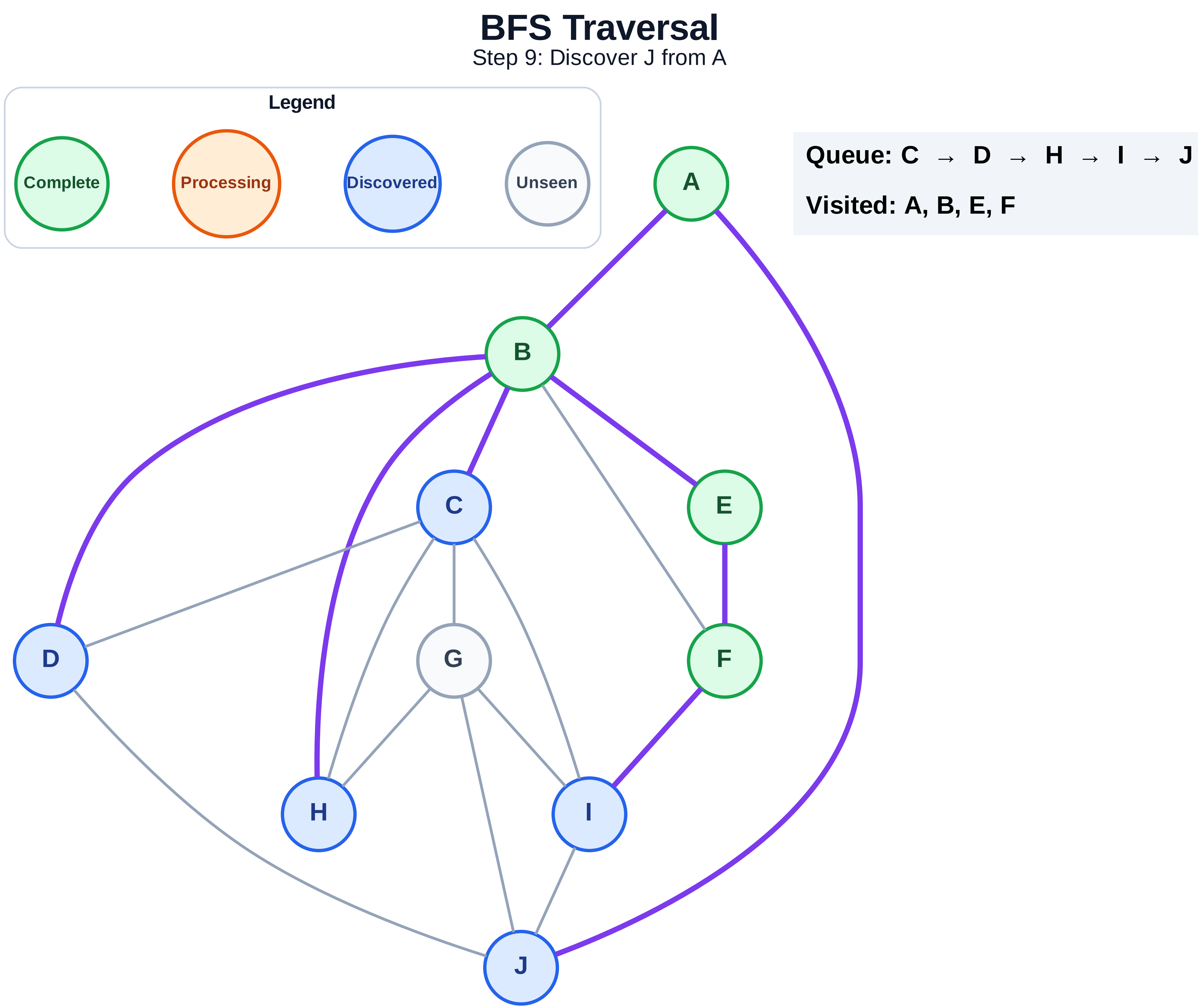
Step 7: Discover I from F



BFS Traversal

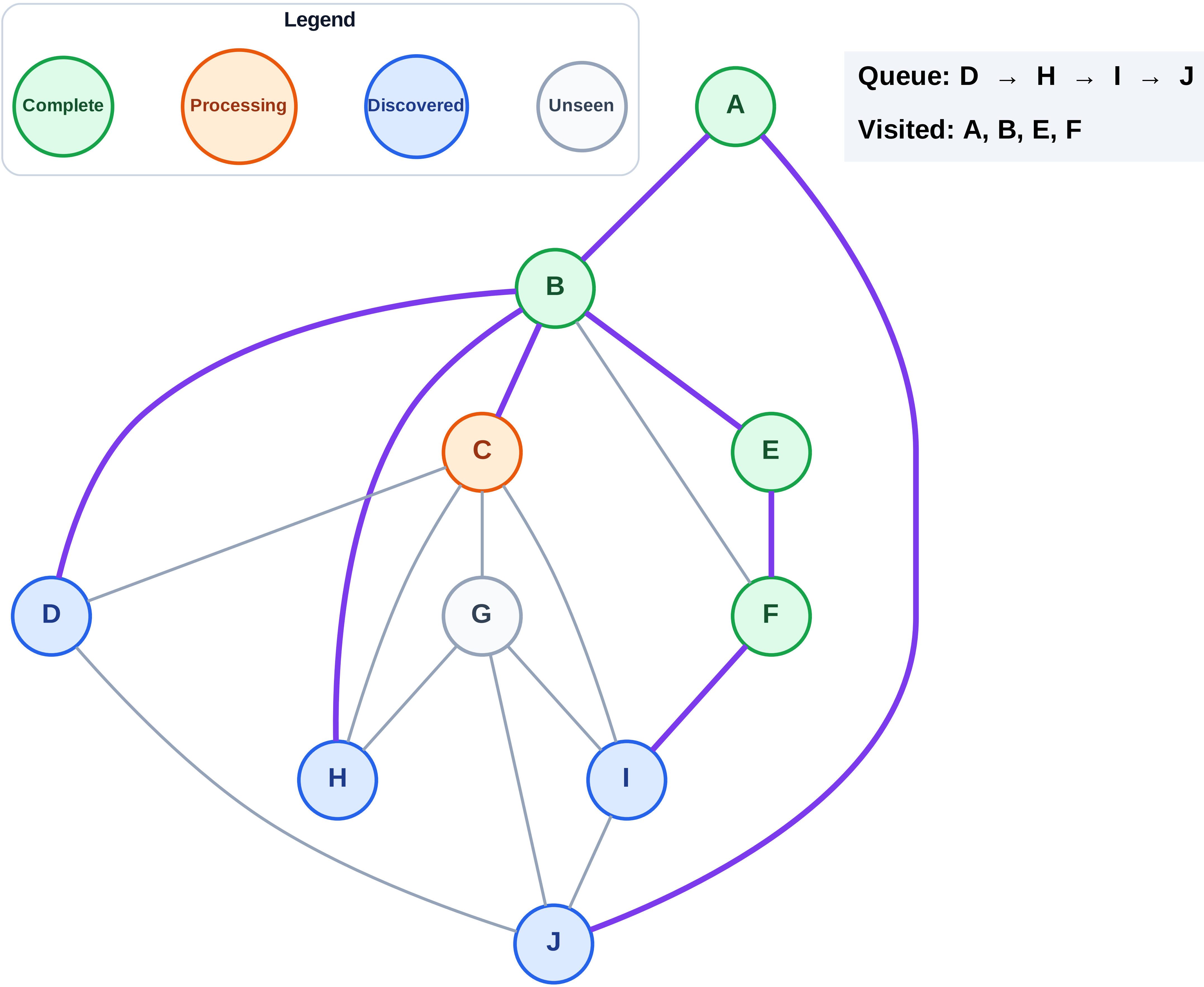
Step 8: Process node A

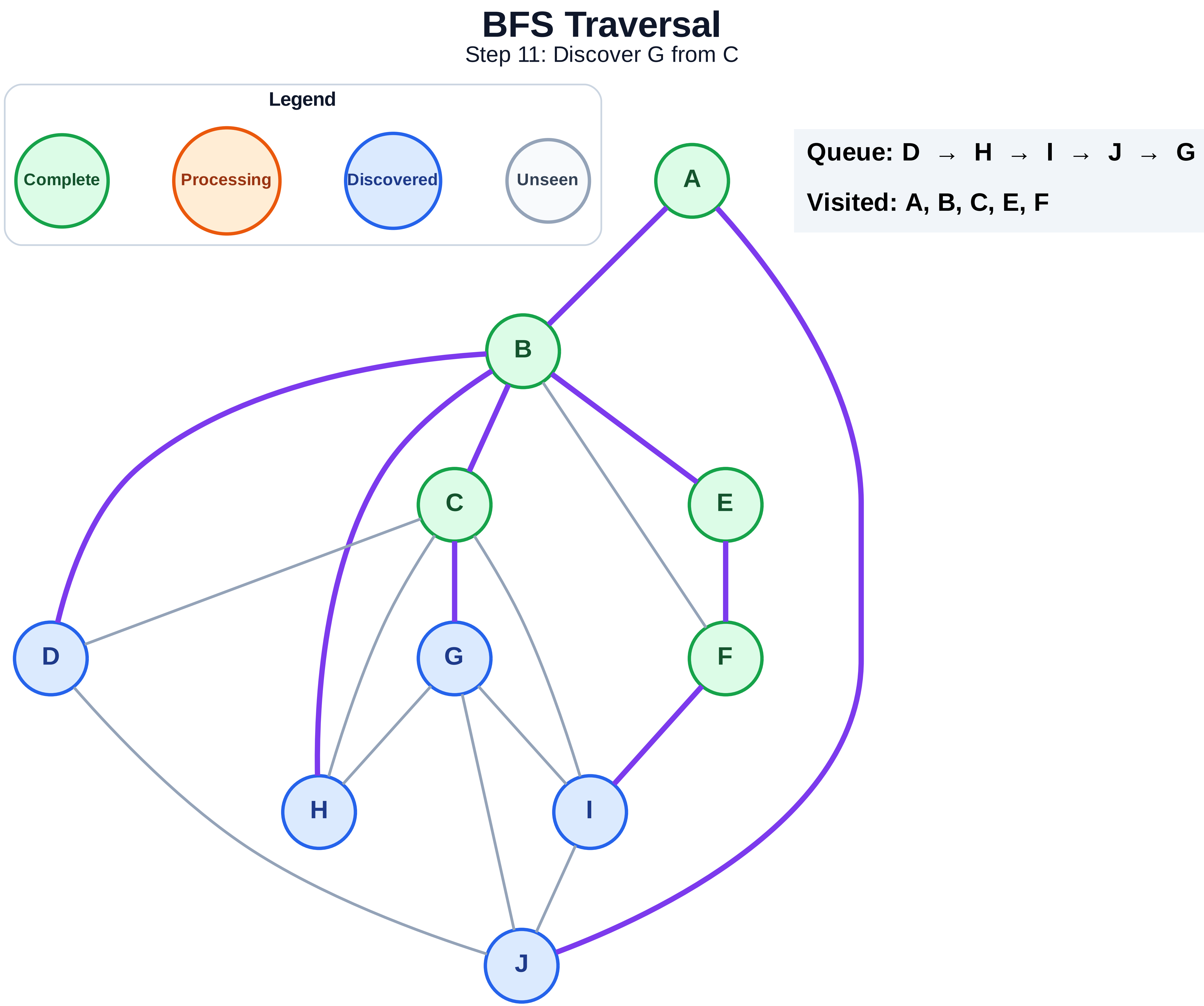



BFS Traversal
Step 9: Discover J from A
Legend
Complete
Processing
Discovered
Unseen
Queue: C → D → H → I → J
Visited: A, B, E, F
A
B
C
D
E
F
G
H
I
J

BFS Traversal

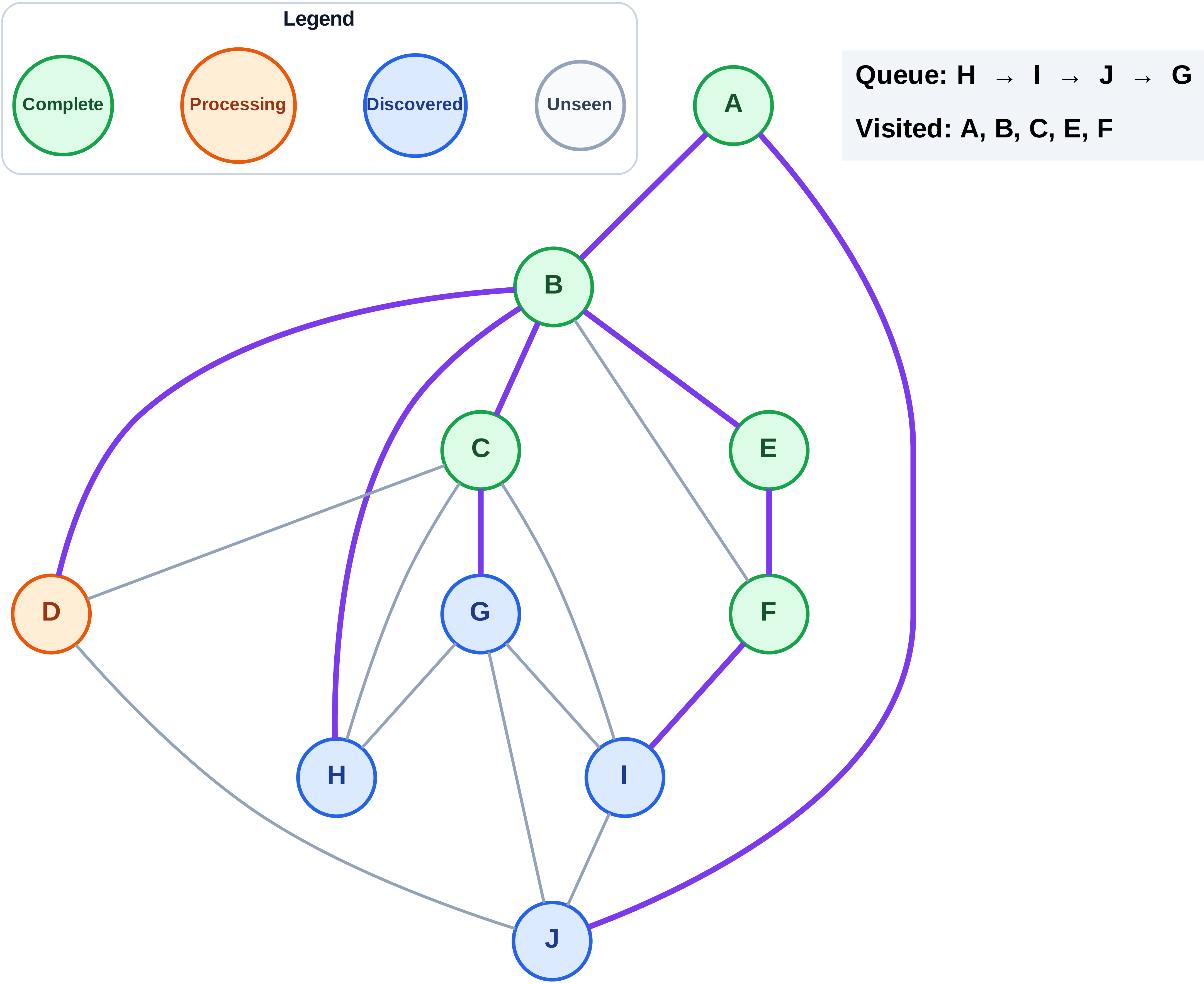
Step 10: Process node C




BFS Traversal
Step 11: Discover G from C
Legend
Complete
Processing
Discovered
Unseen
Queue: D → H → I → J → G
Visited: A, B, C, E, F
A
B
C
D
E
F
G
H
I
J

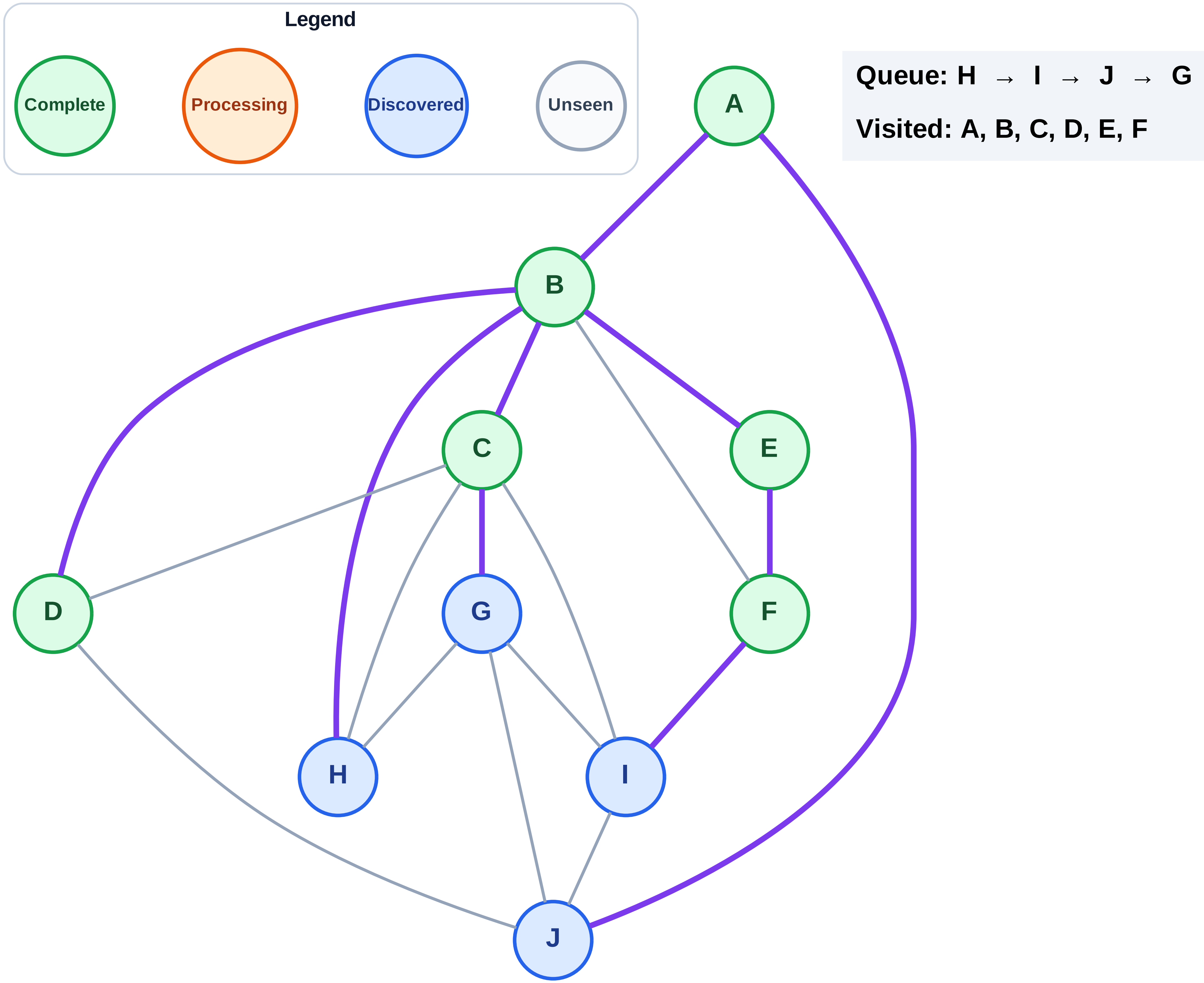
BFS Traversal

Step 12: Process node D



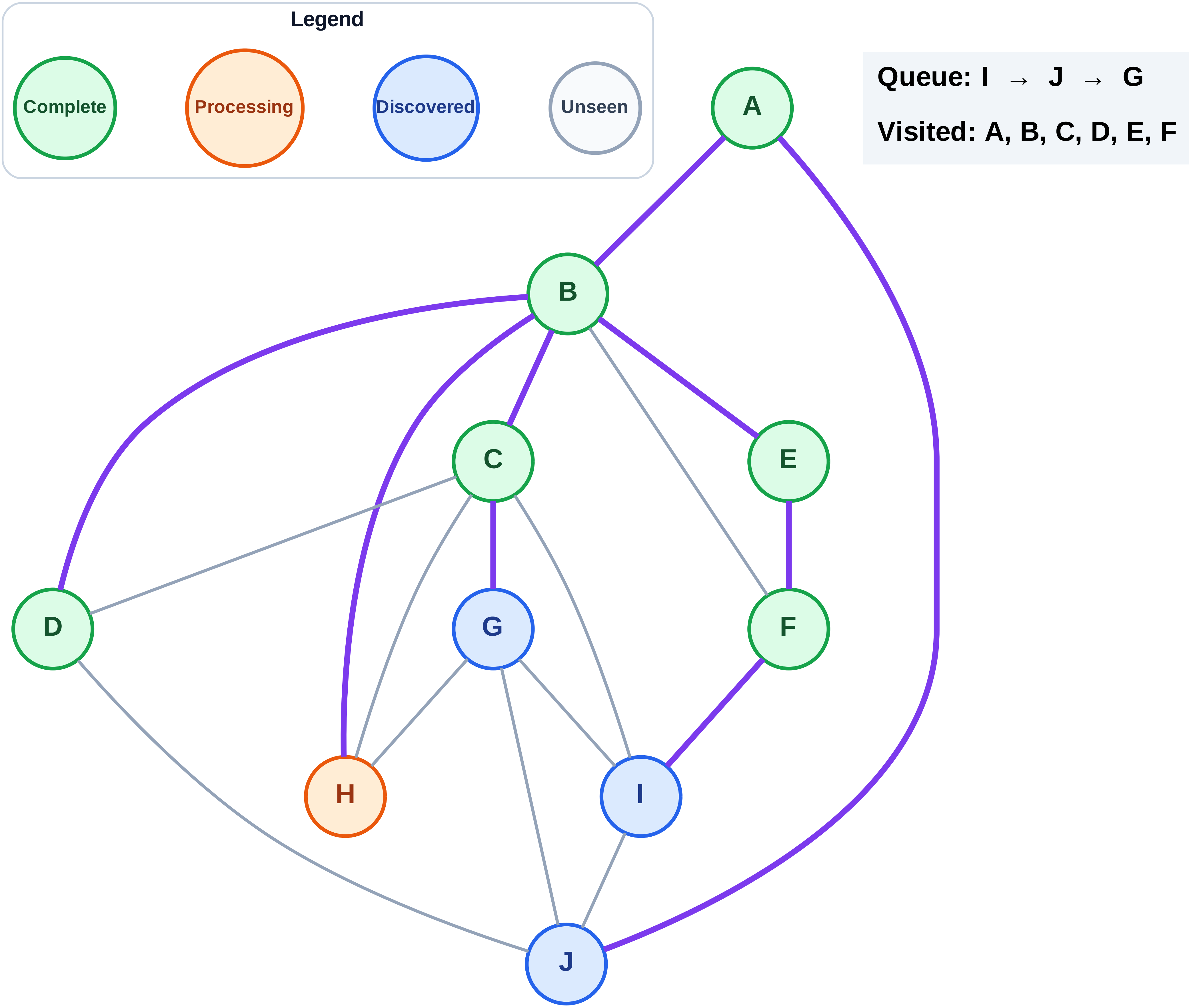
BFS Traversal

Step 13: No new neighbors from D



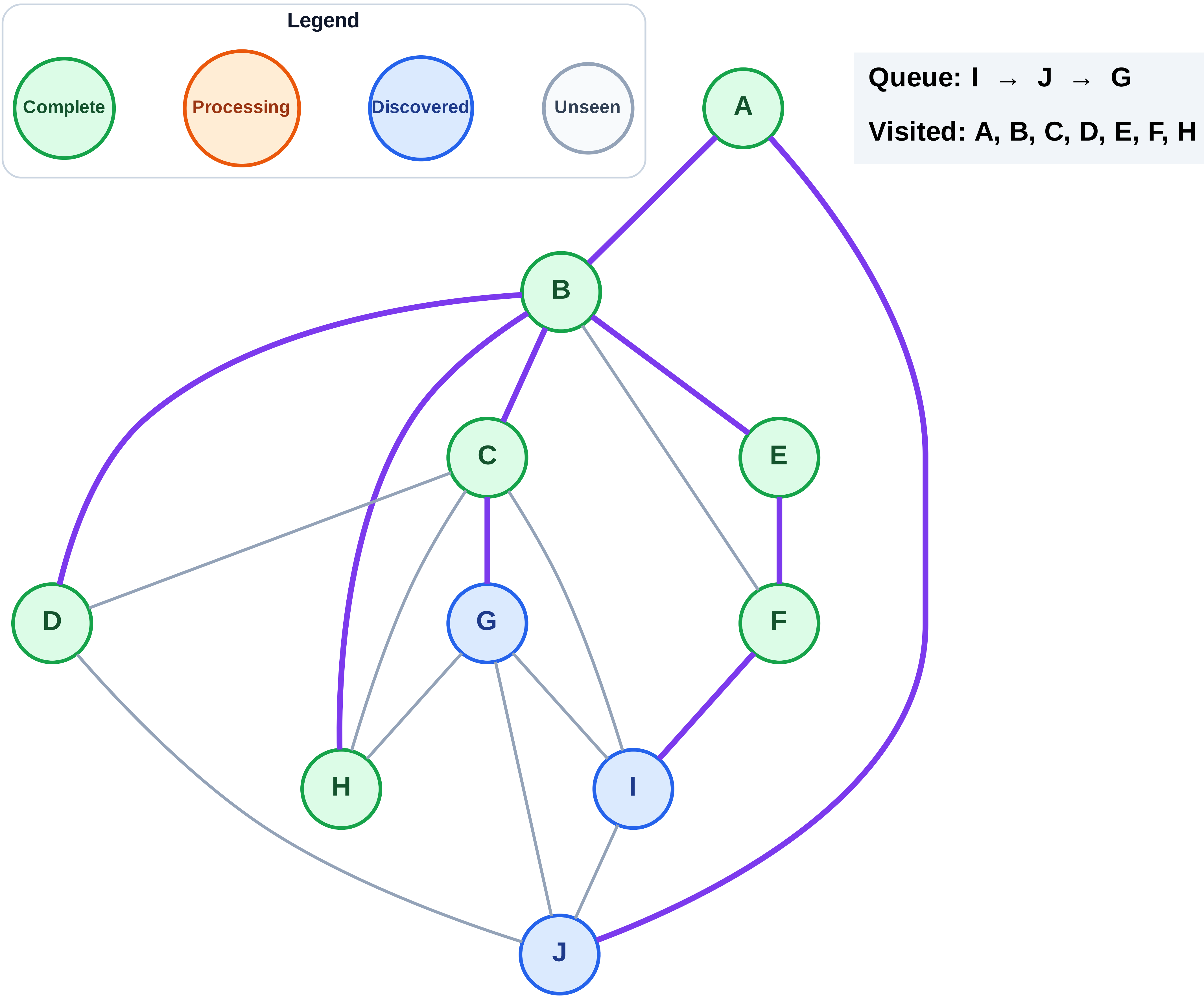
BFS Traversal

Step 14: Process node H



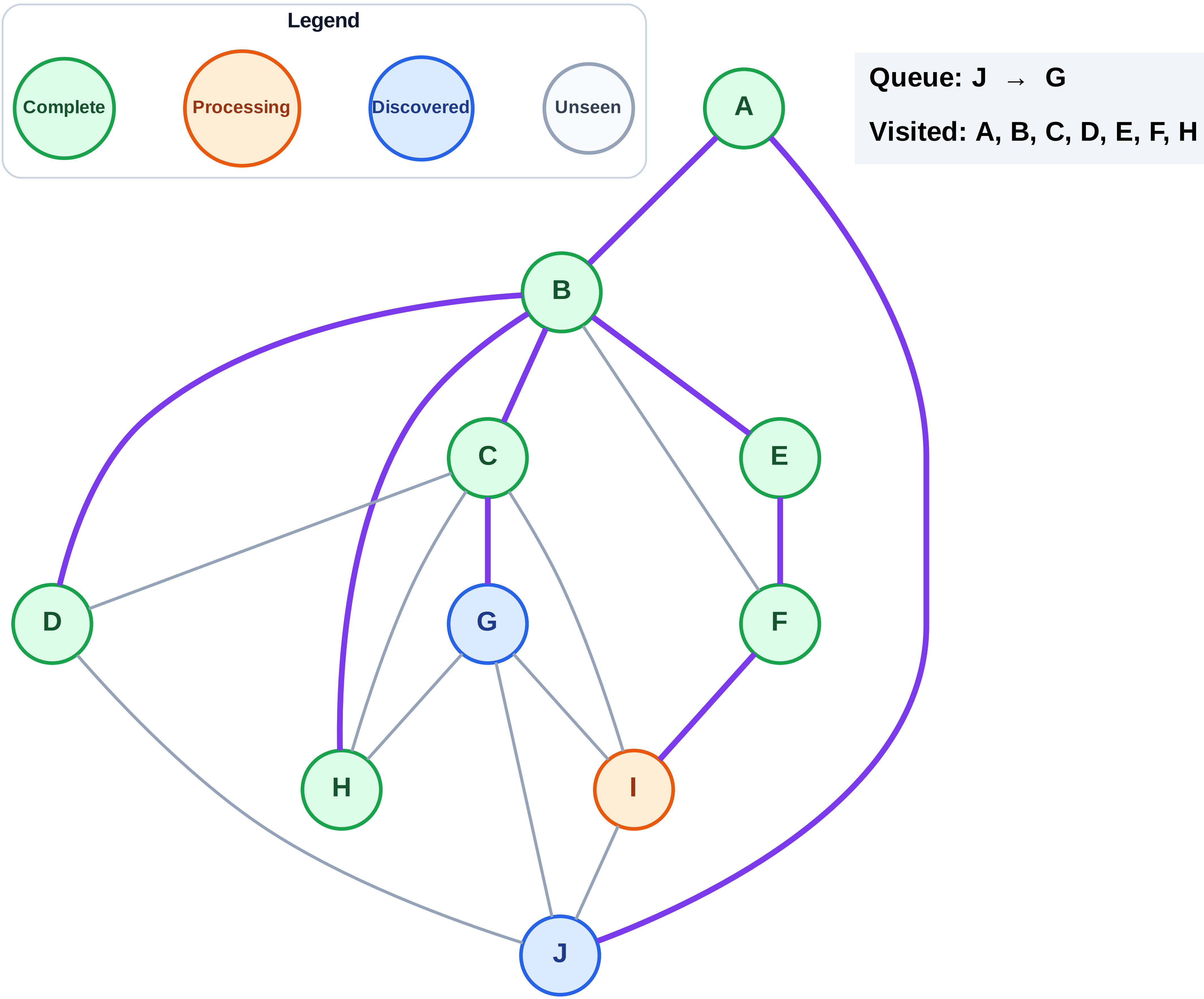
BFS Traversal

Step 15: No new neighbors from H



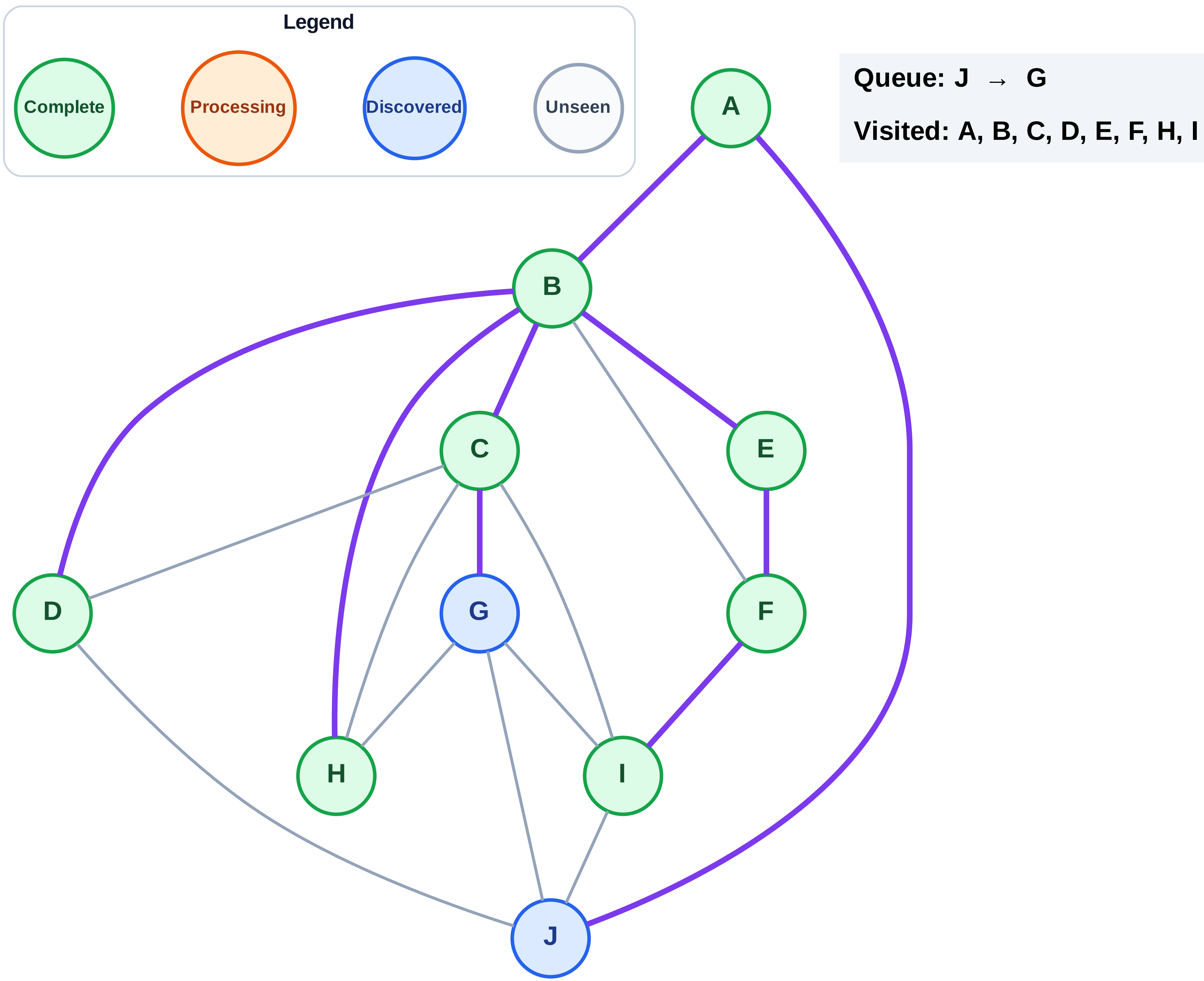
BFS Traversal

Step 16: Process node I



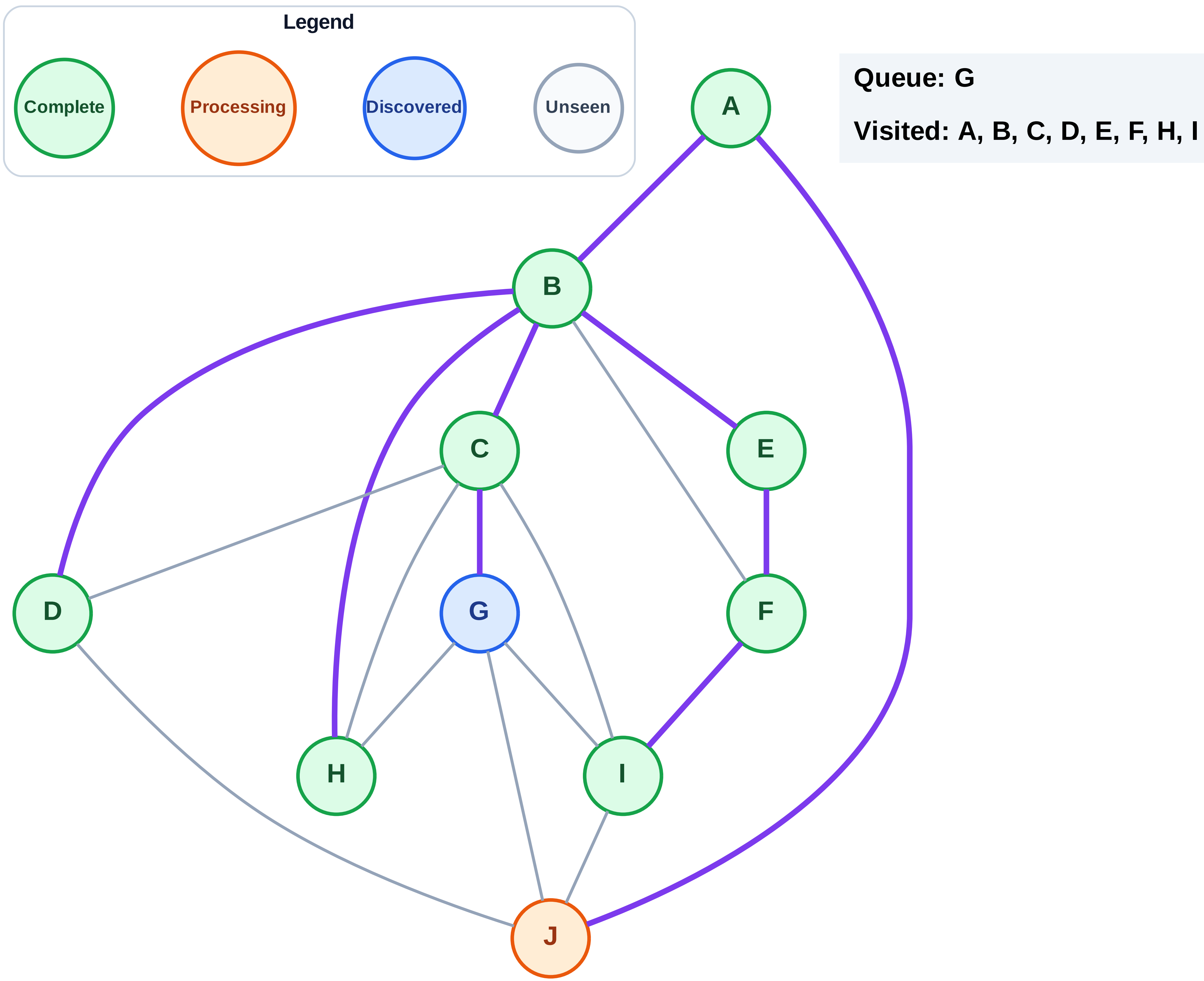
BFS Traversal

Step 17: No new neighbors from I



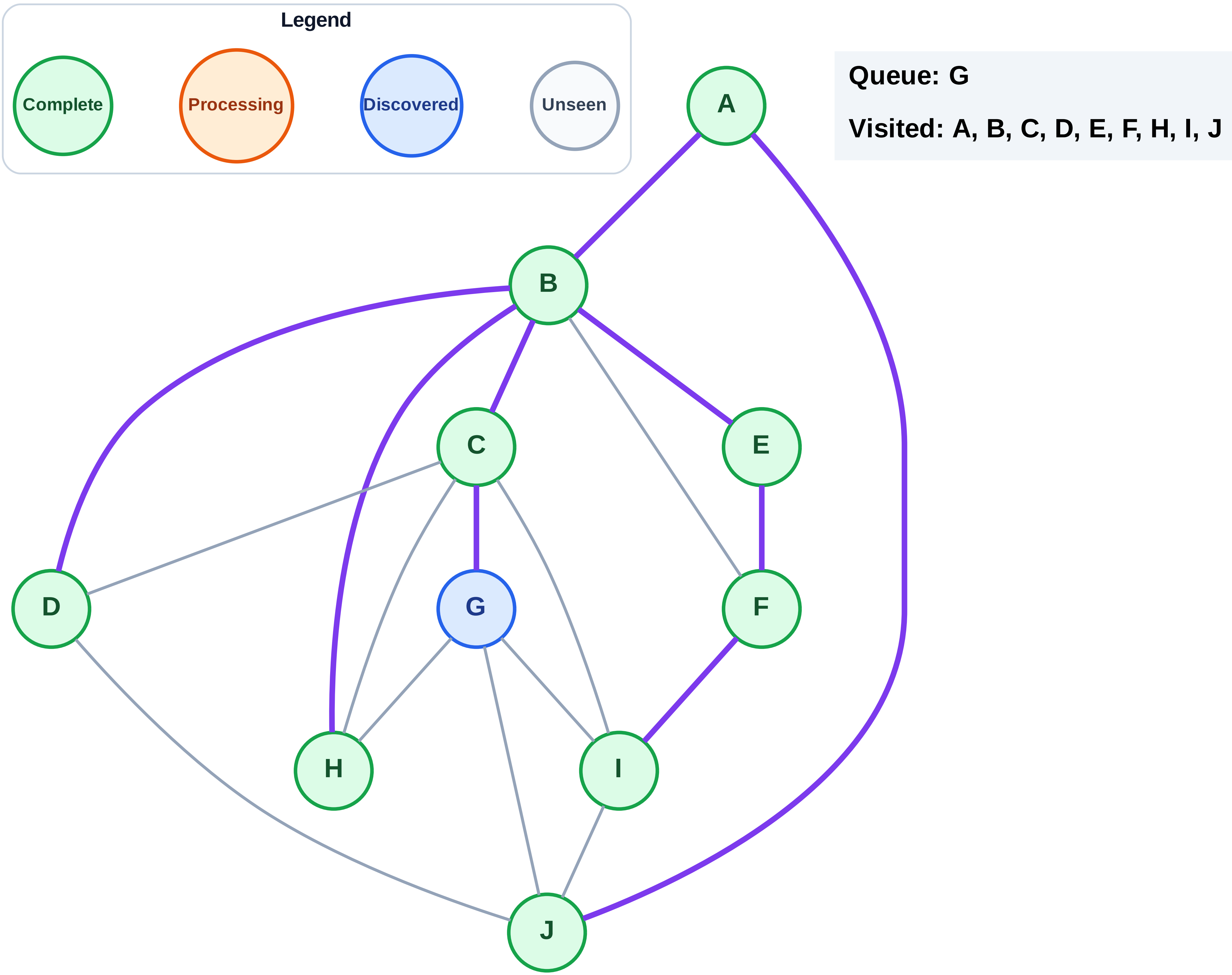
BFS Traversal

Step 18: Process node J



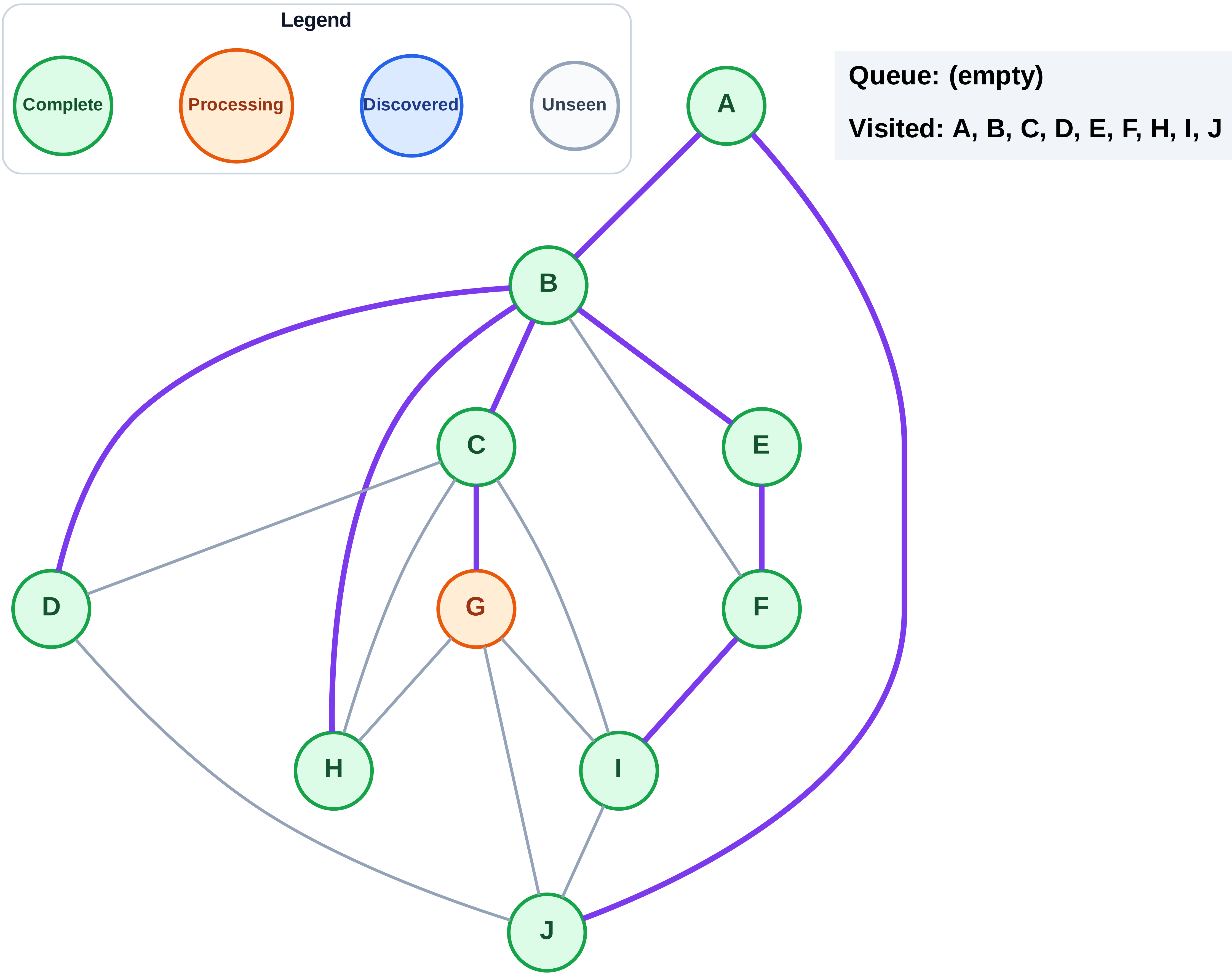
BFS Traversal

Step 19: No new neighbors from J



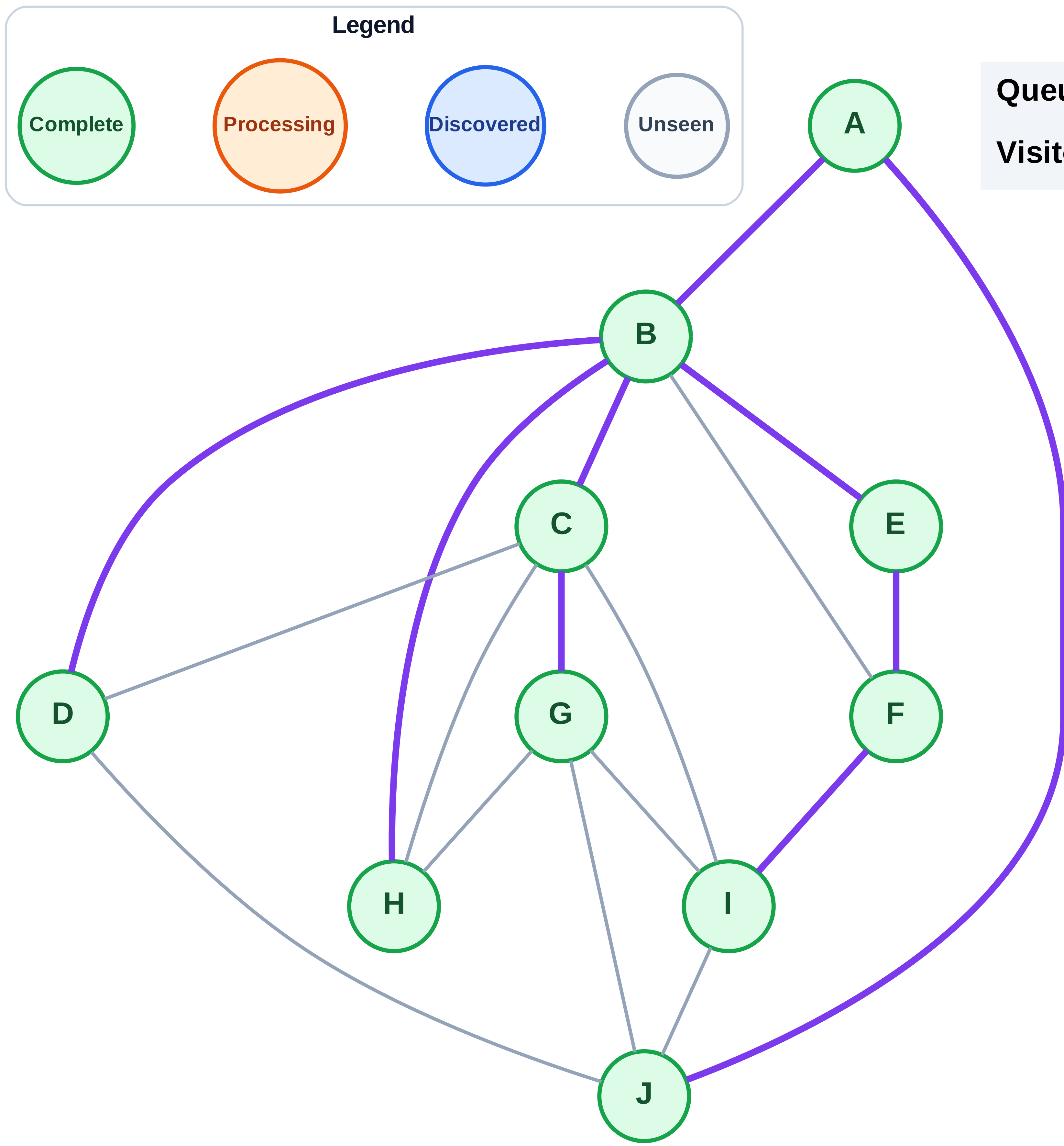
BFS Traversal

Step 20: Process node G



BFS Traversal

Step 21: No new neighbors from G

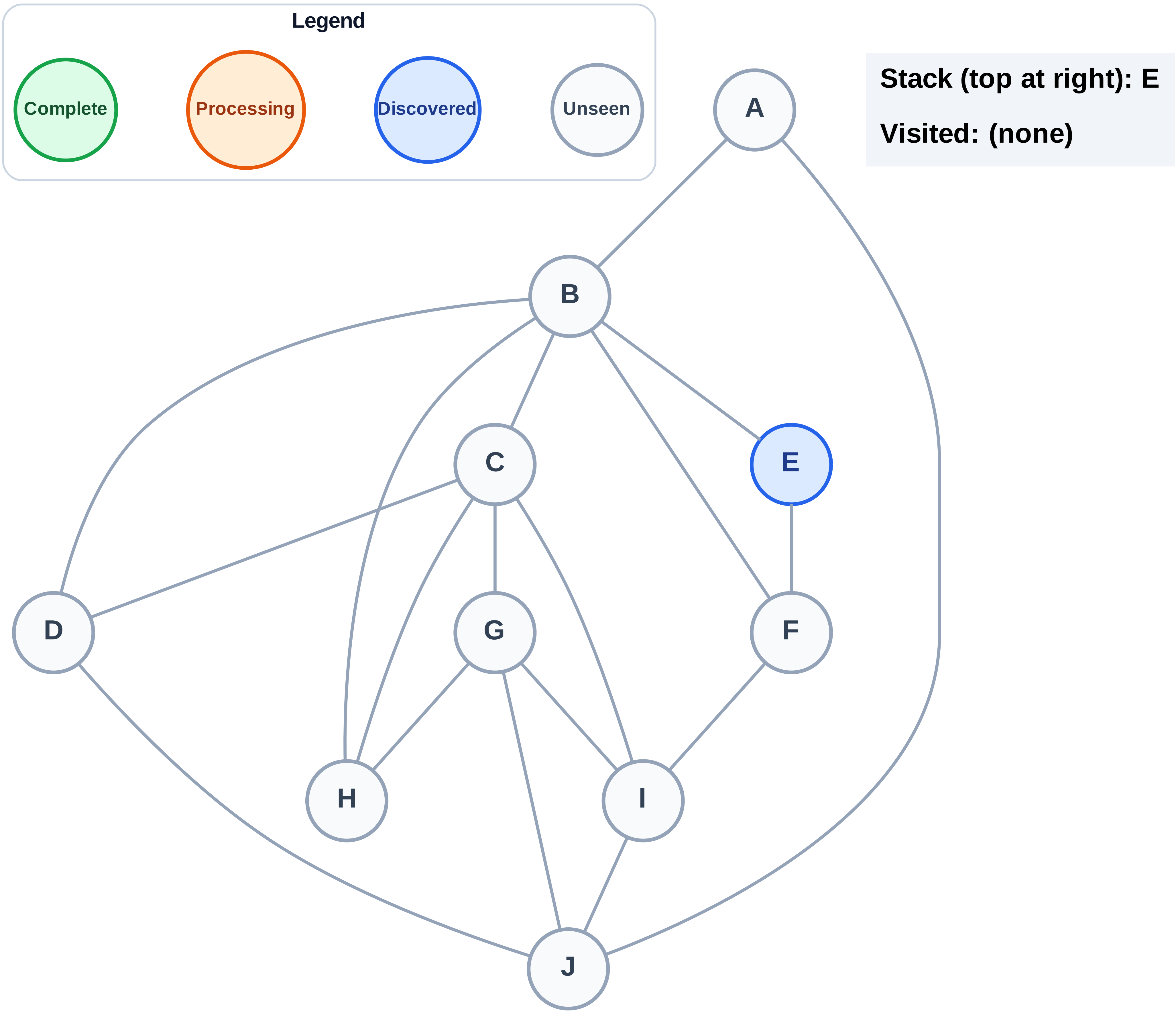


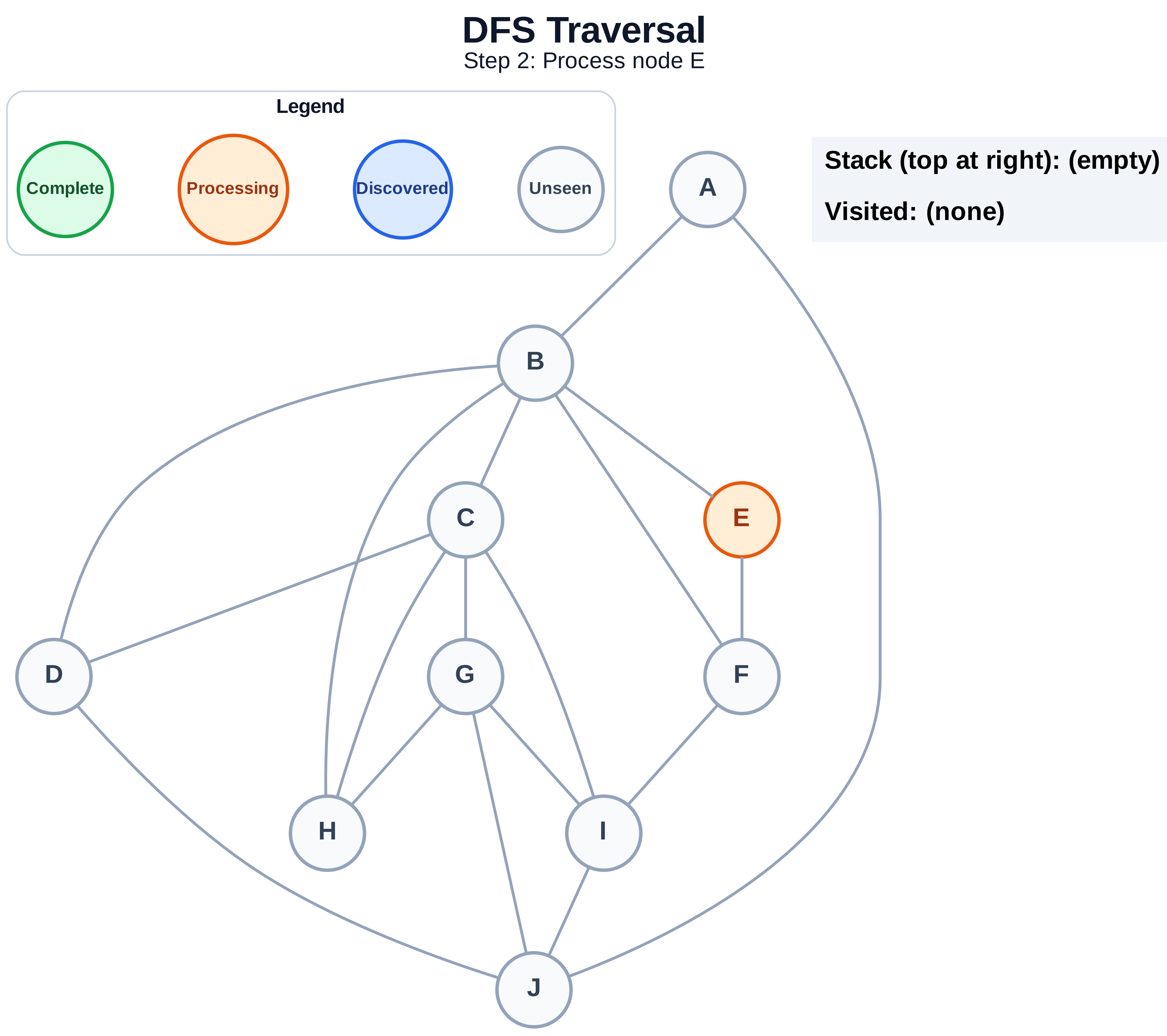
Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

DFS Traversal

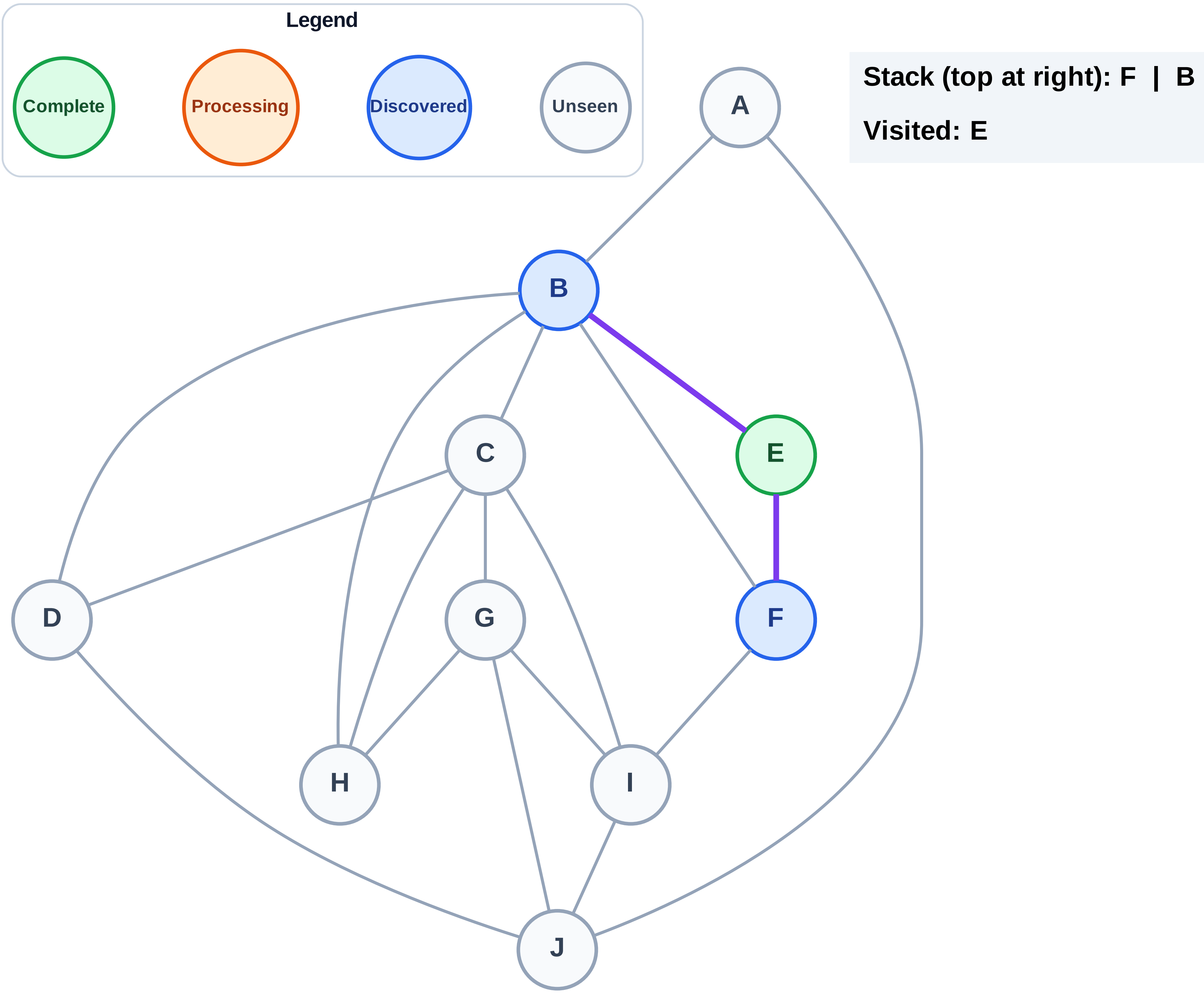
Step 1: Start at E




DFS Traversal
Step 2: Process node E
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: (none)
A
B
C
D
E
F
G
H
I
J

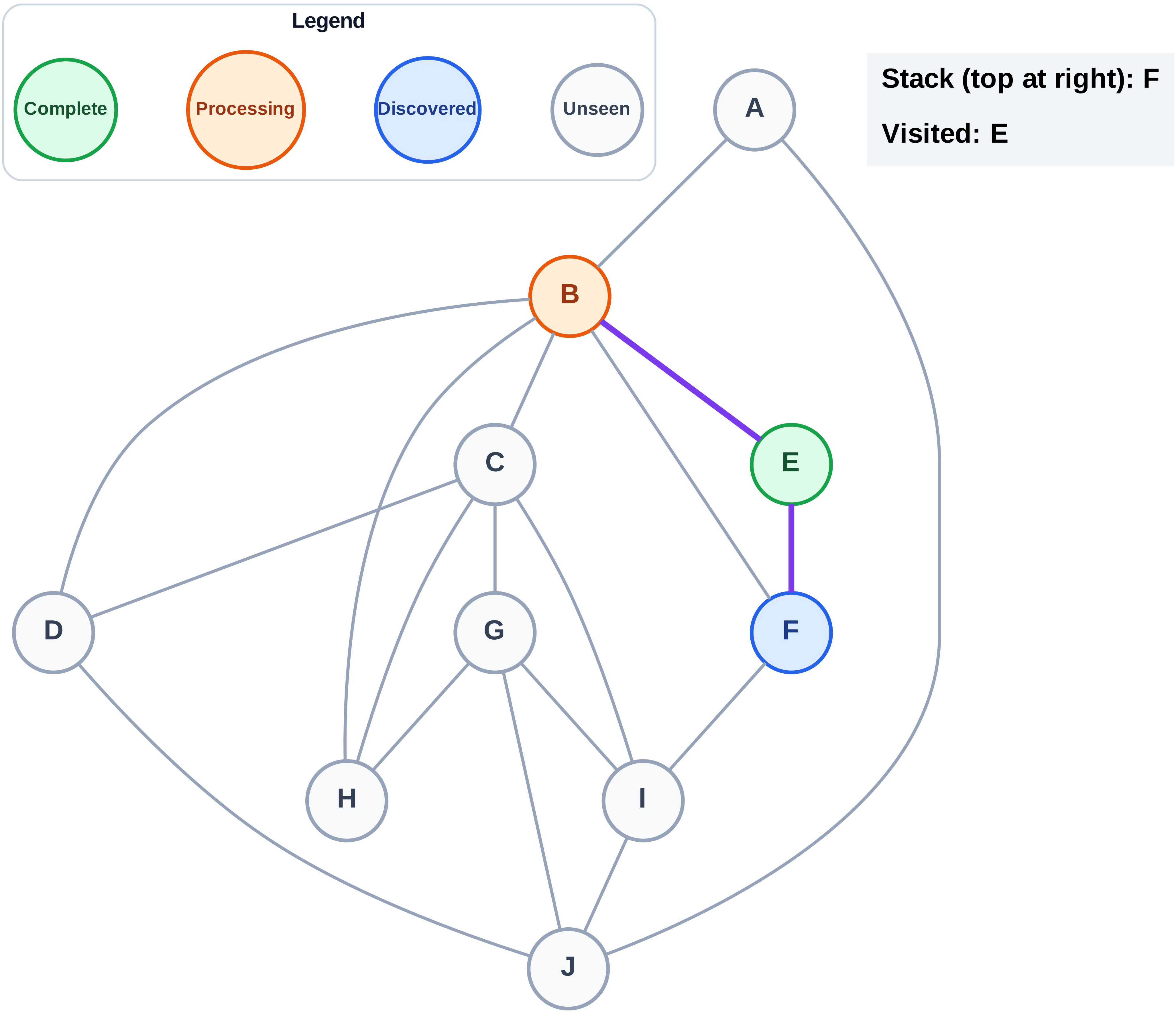
DFS Traversal

Step 3: Discover F, B from E



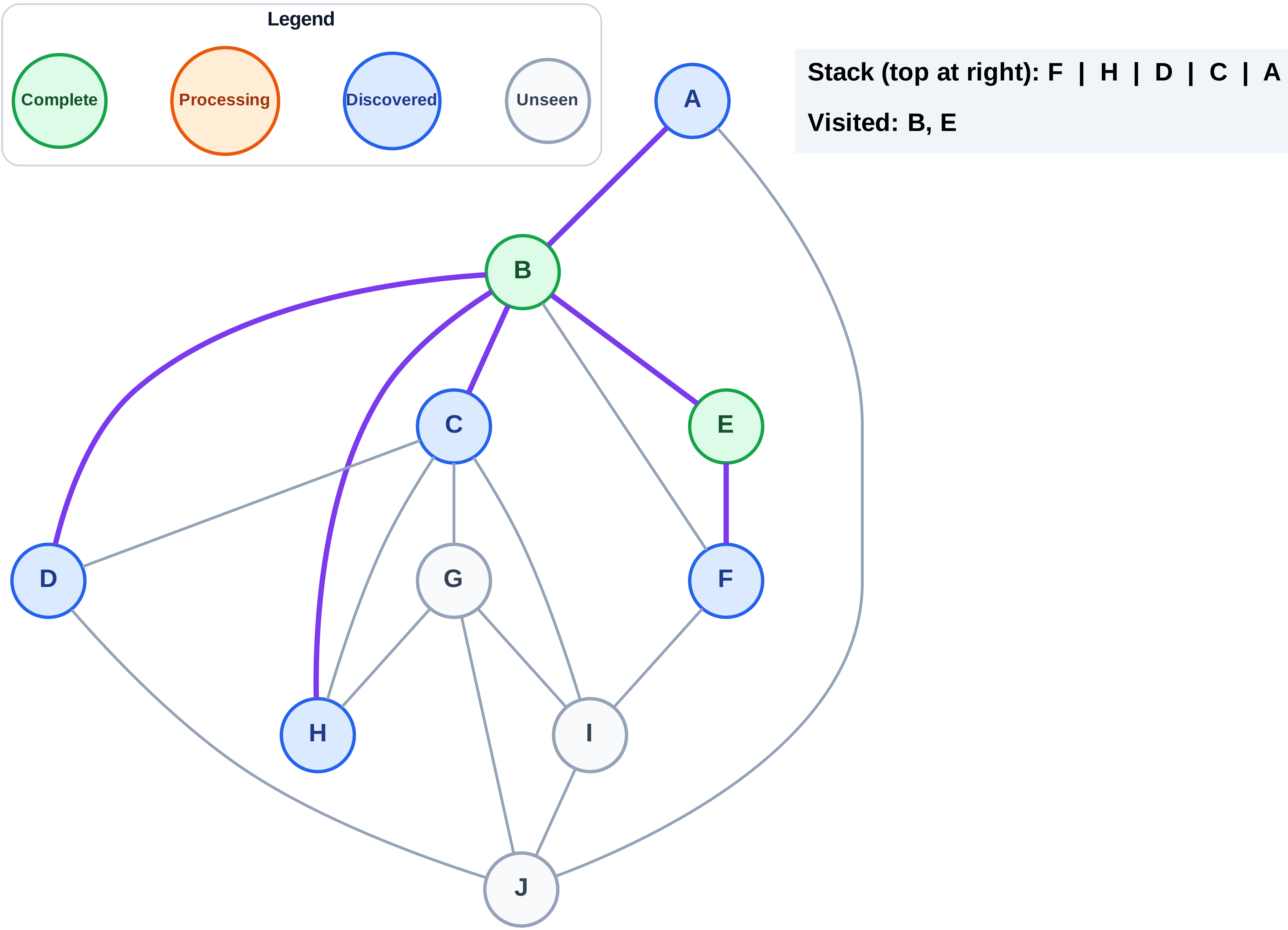
DFS Traversal

Step 4: Process node B



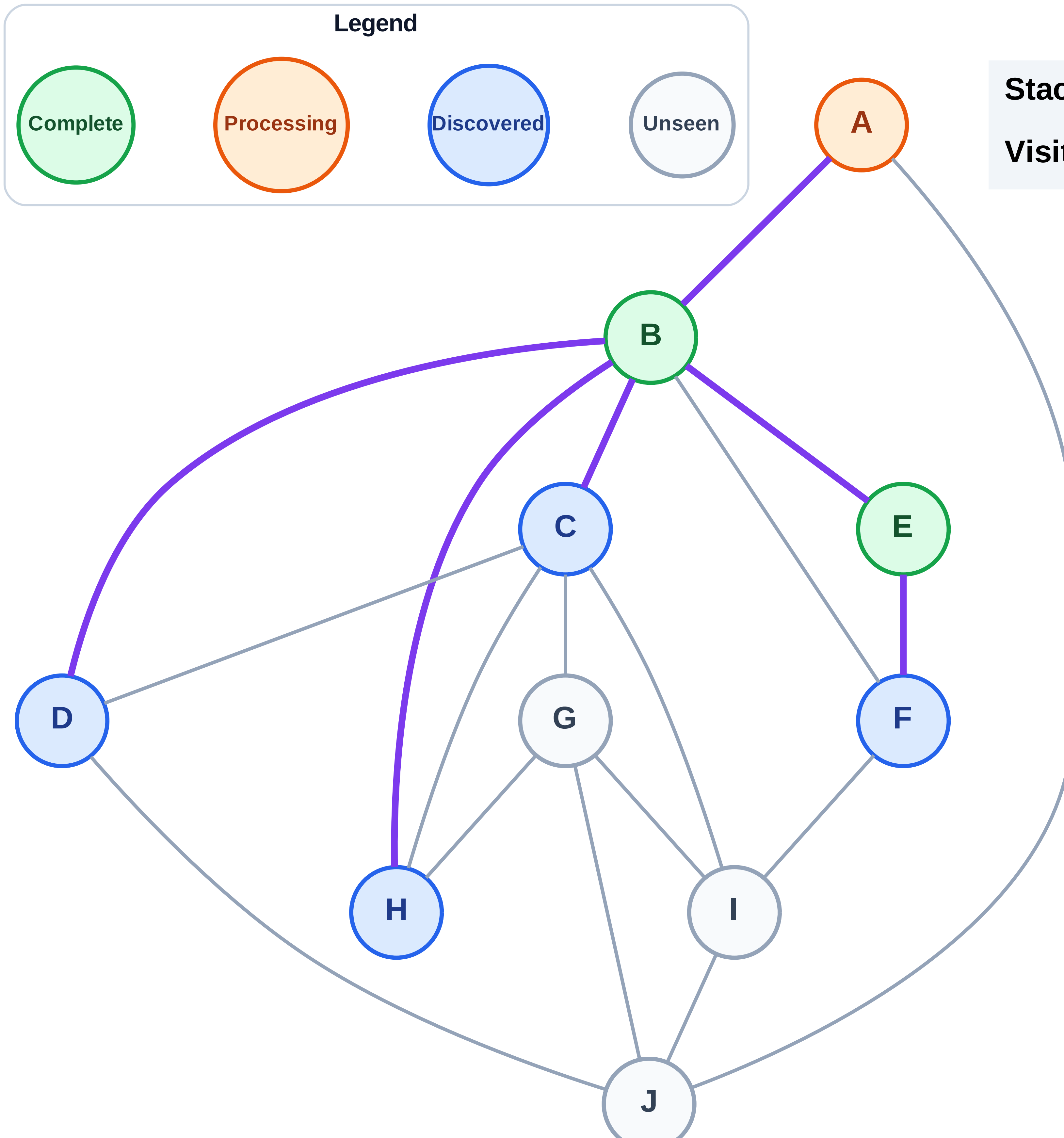
DFS Traversal

Step 5: Discover H, D, C, A from B



DFS Traversal

Step 6: Process node A

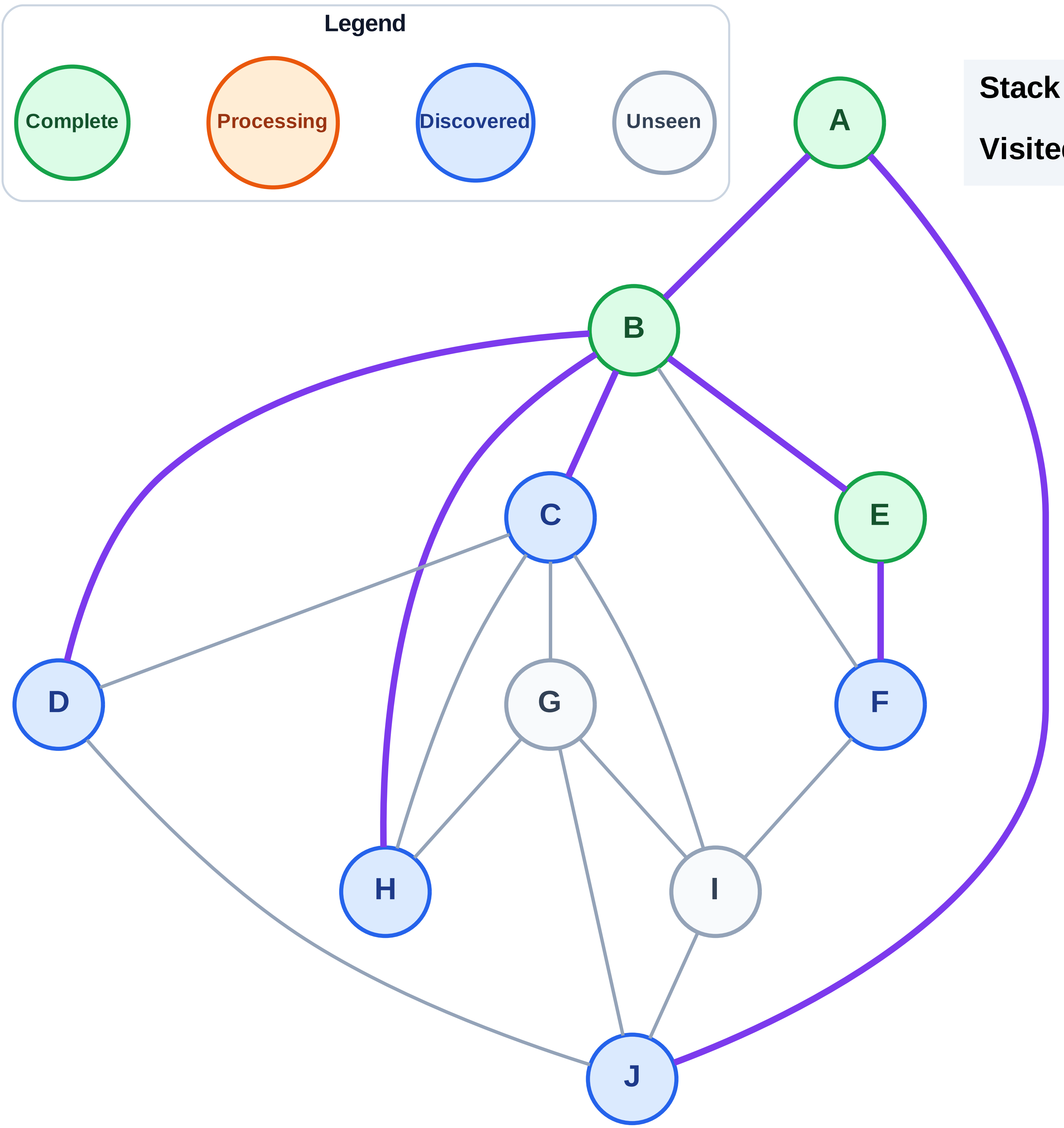


Stack (top at right): F | H | D | C

Visited: B, E

DFS Traversal

Step 7: Discover J from A

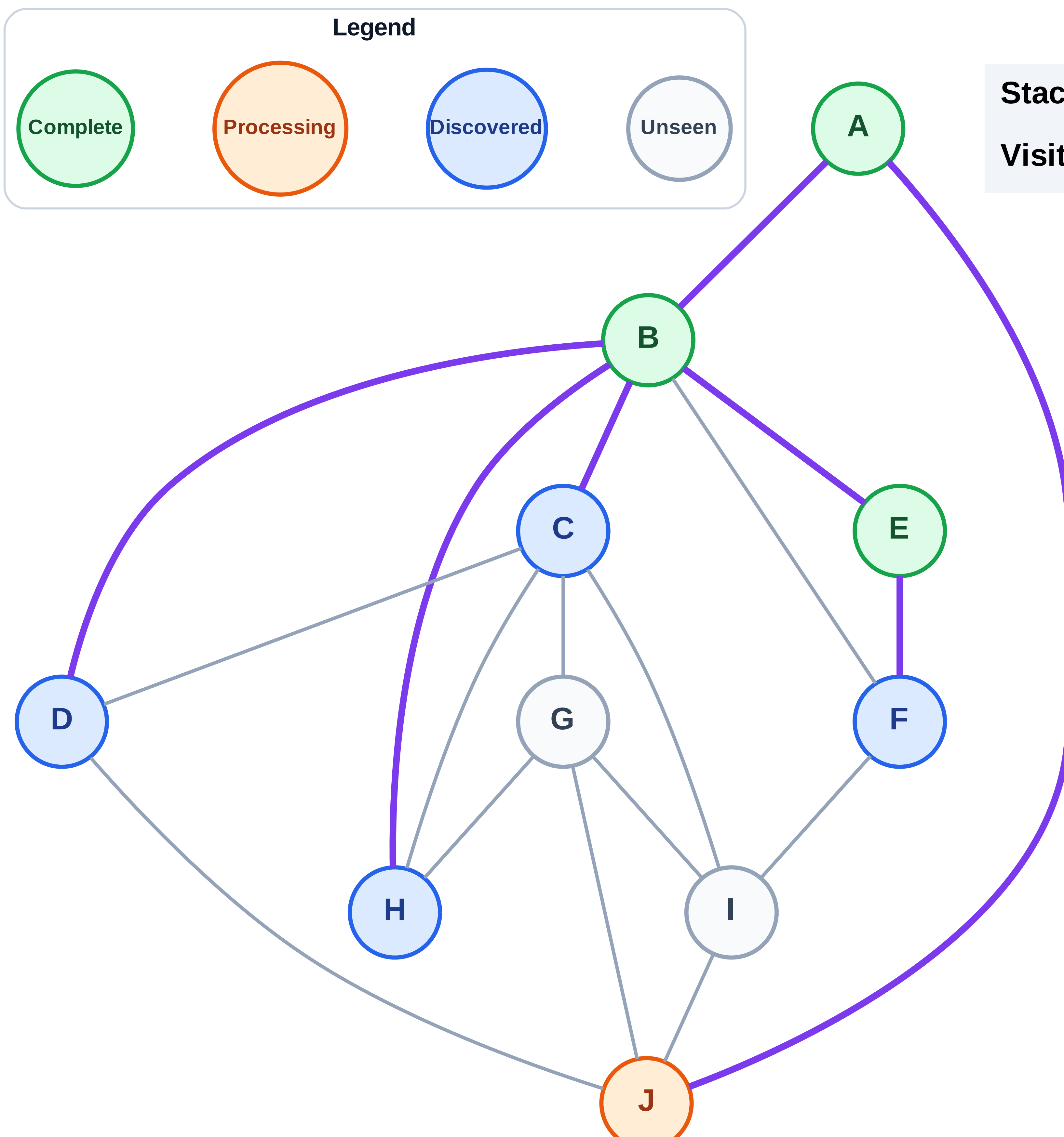


Stack (top at right): F | H | D | C | J

Visited: A, B, E

DFS Traversal

Step 8: Process node J

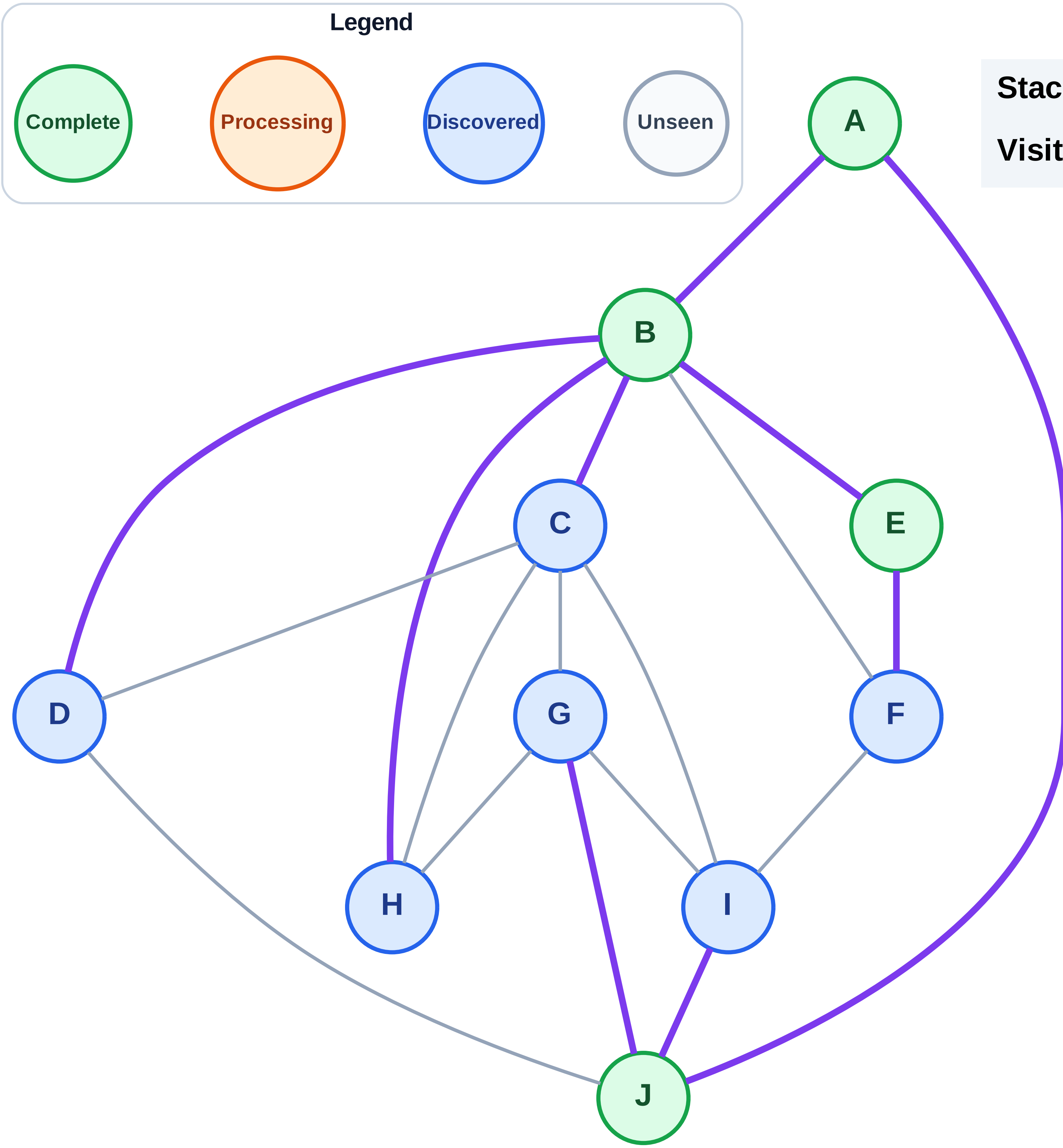


Stack (top at right): F | H | D | C

Visited: A, B, E

DFS Traversal

Step 9: Discover I, G from J

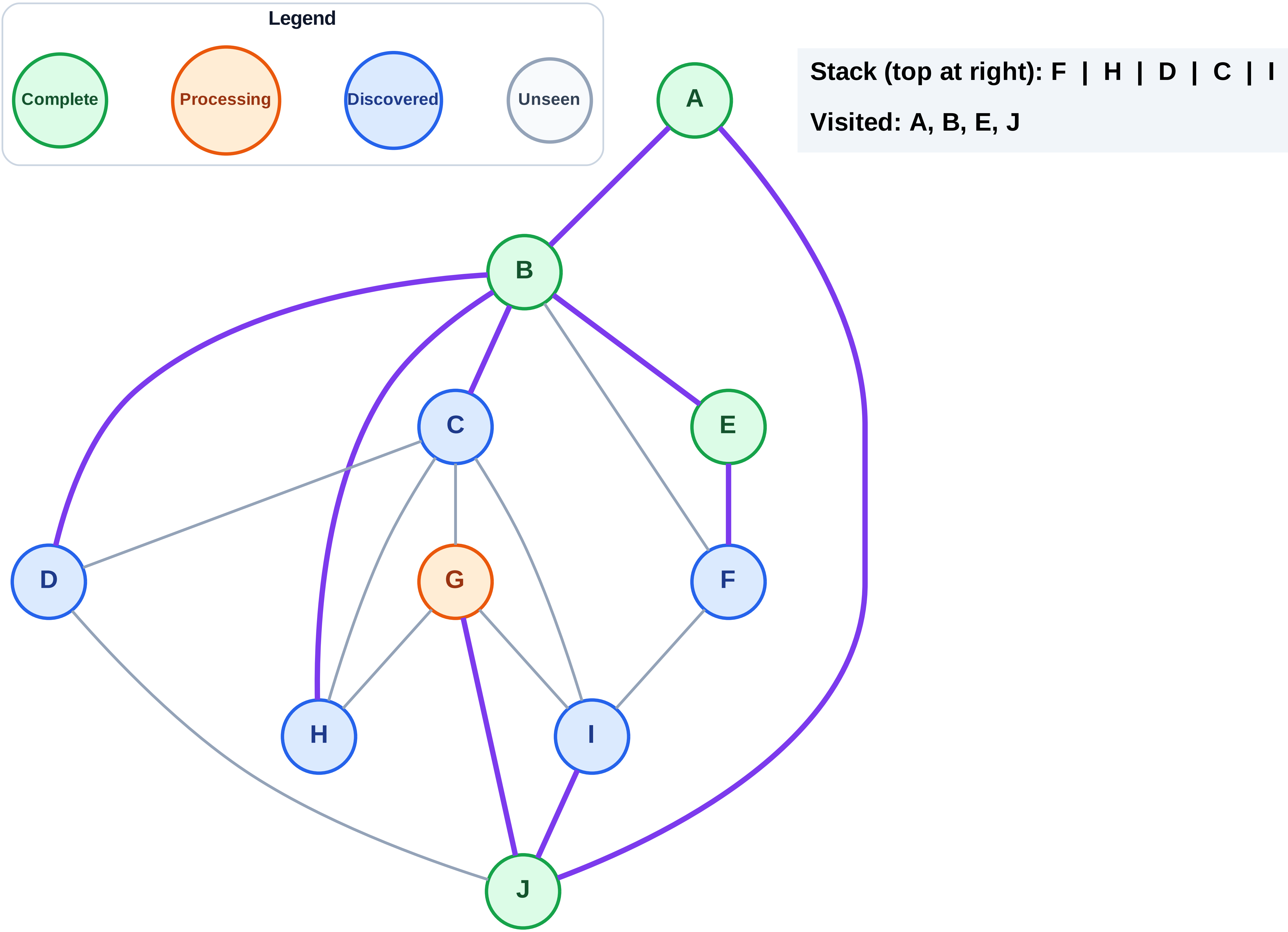


Stack (top at right): F | H | D | C | I | G

Visited: A, B, E, J

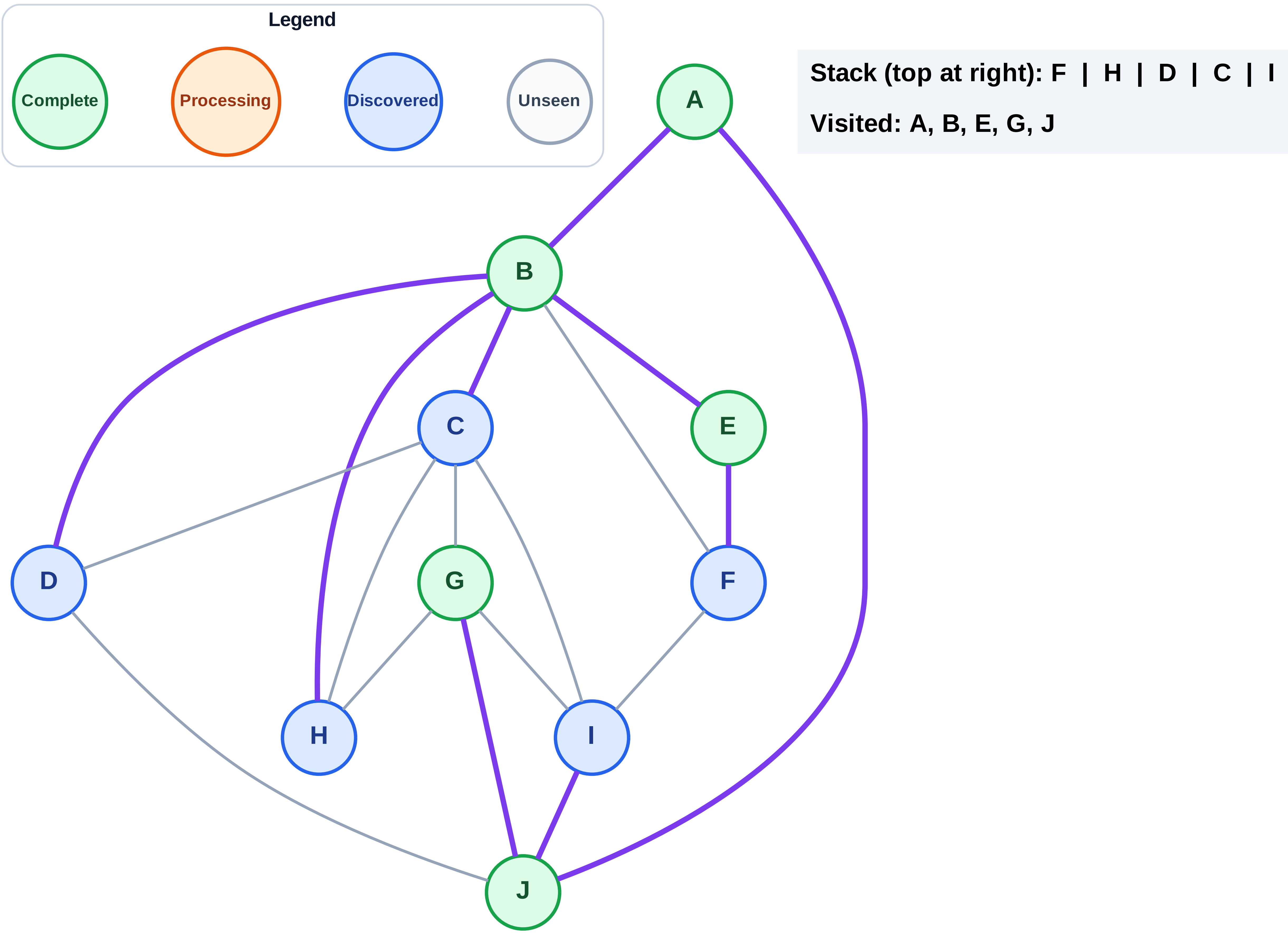
DFS Traversal

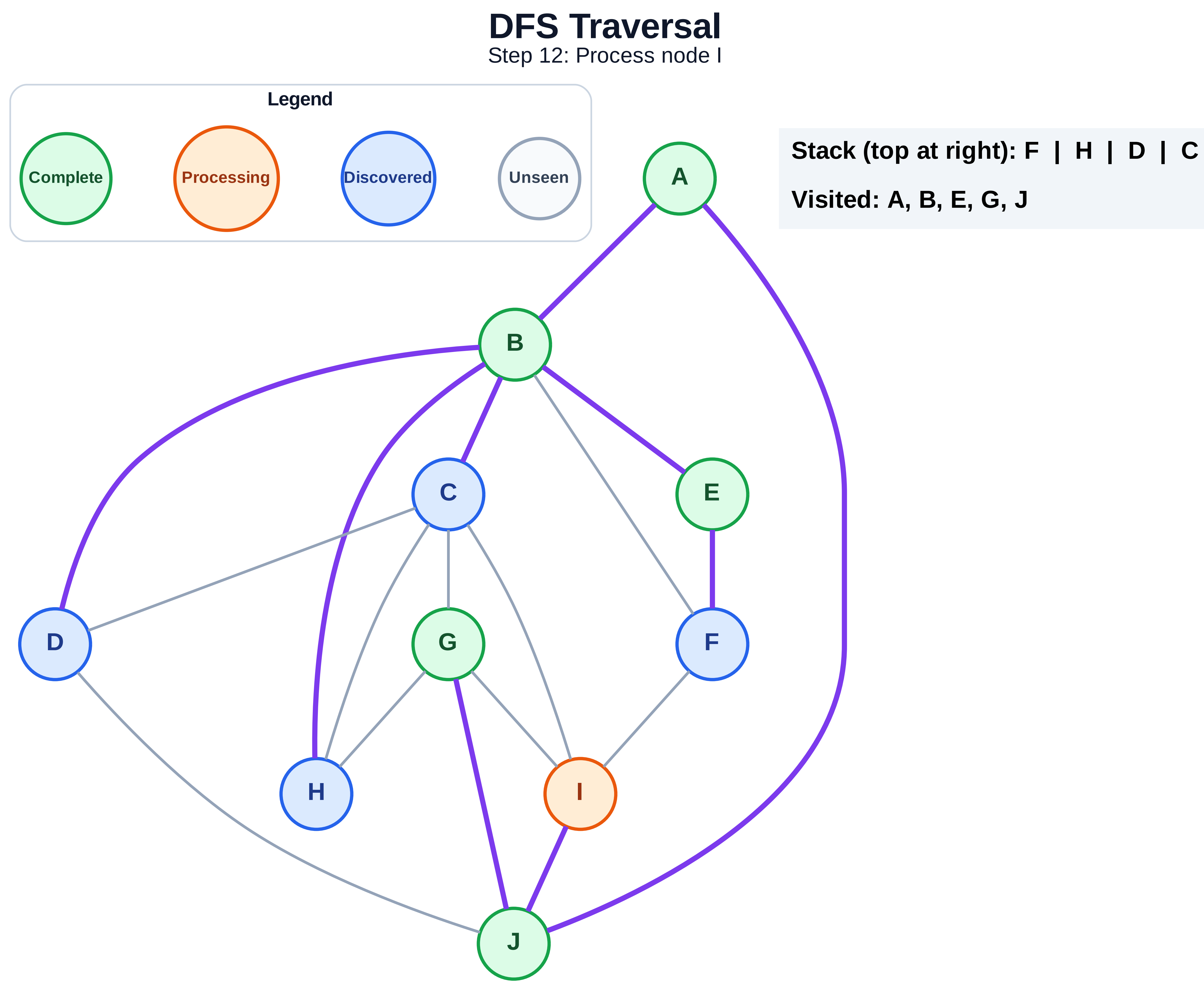
Step 10: Process node G



DFS Traversal

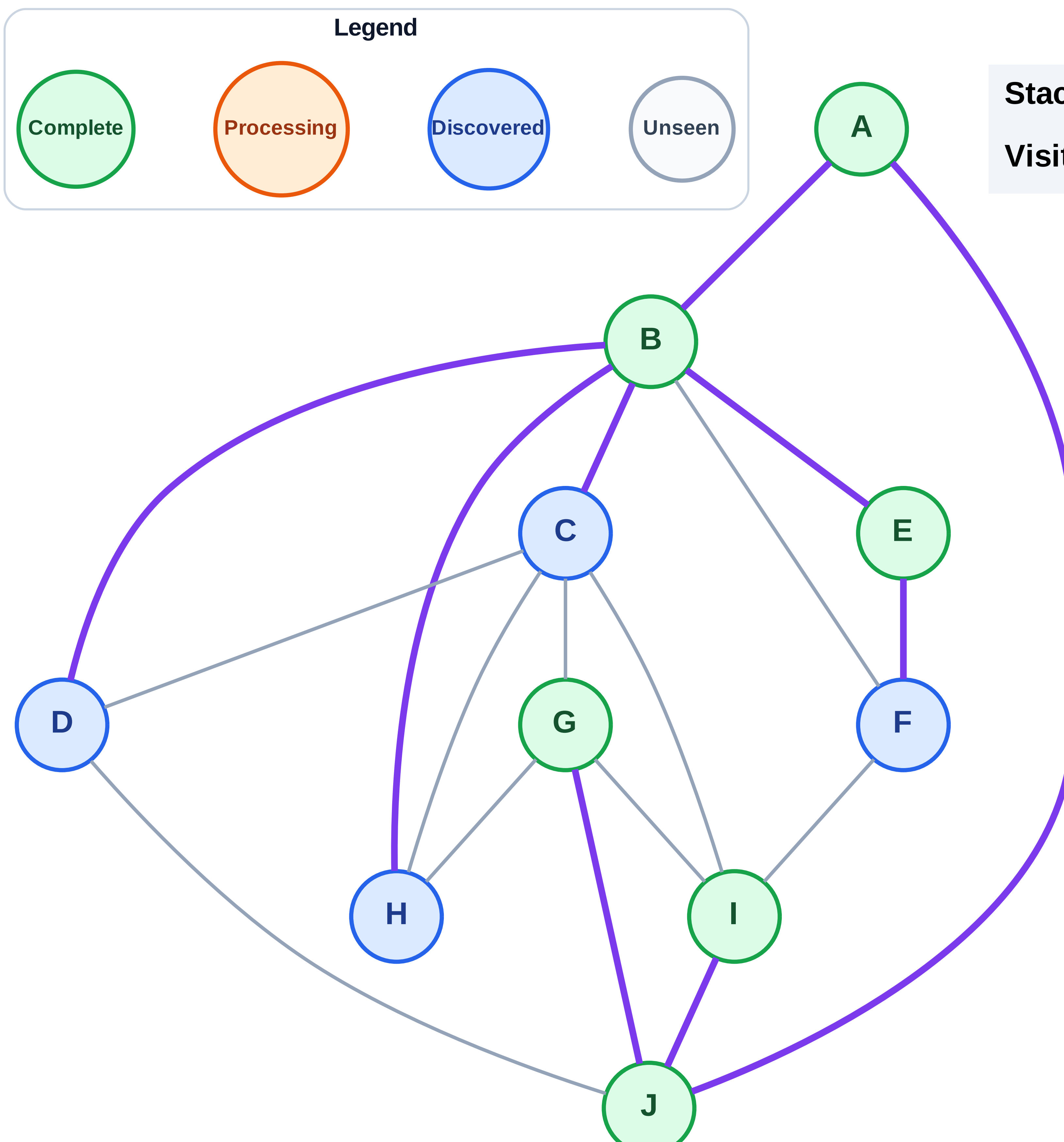
Step 11: No new neighbors from G




DFS Traversal
Step 12: Process node I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): F | H | D | C
Visited: A, B, E, G, J
A
B
C
D
E
F
G
H
I
J

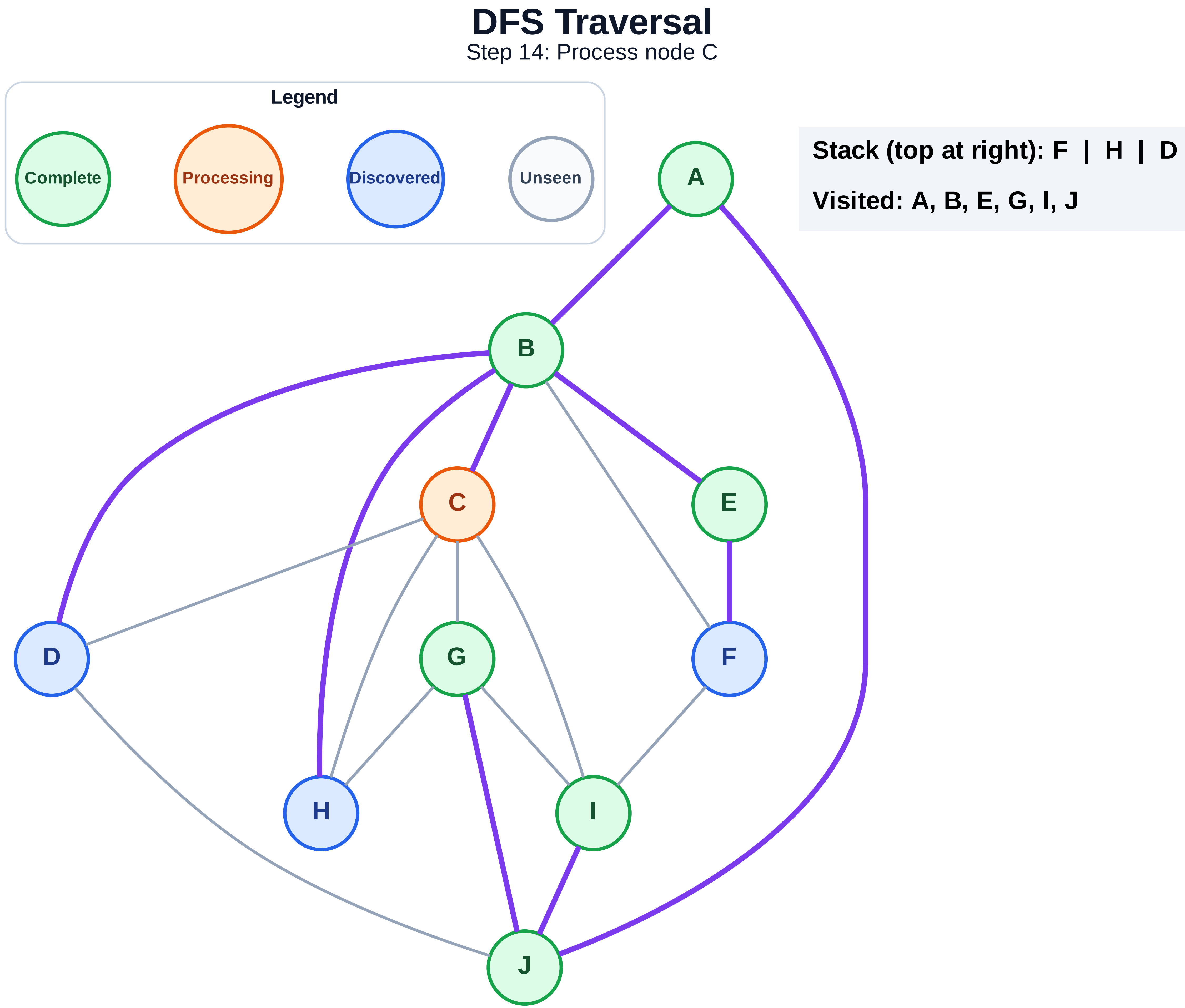
DFS Traversal

Step 13: No new neighbors from I



Stack (top at right): F | H | D | C

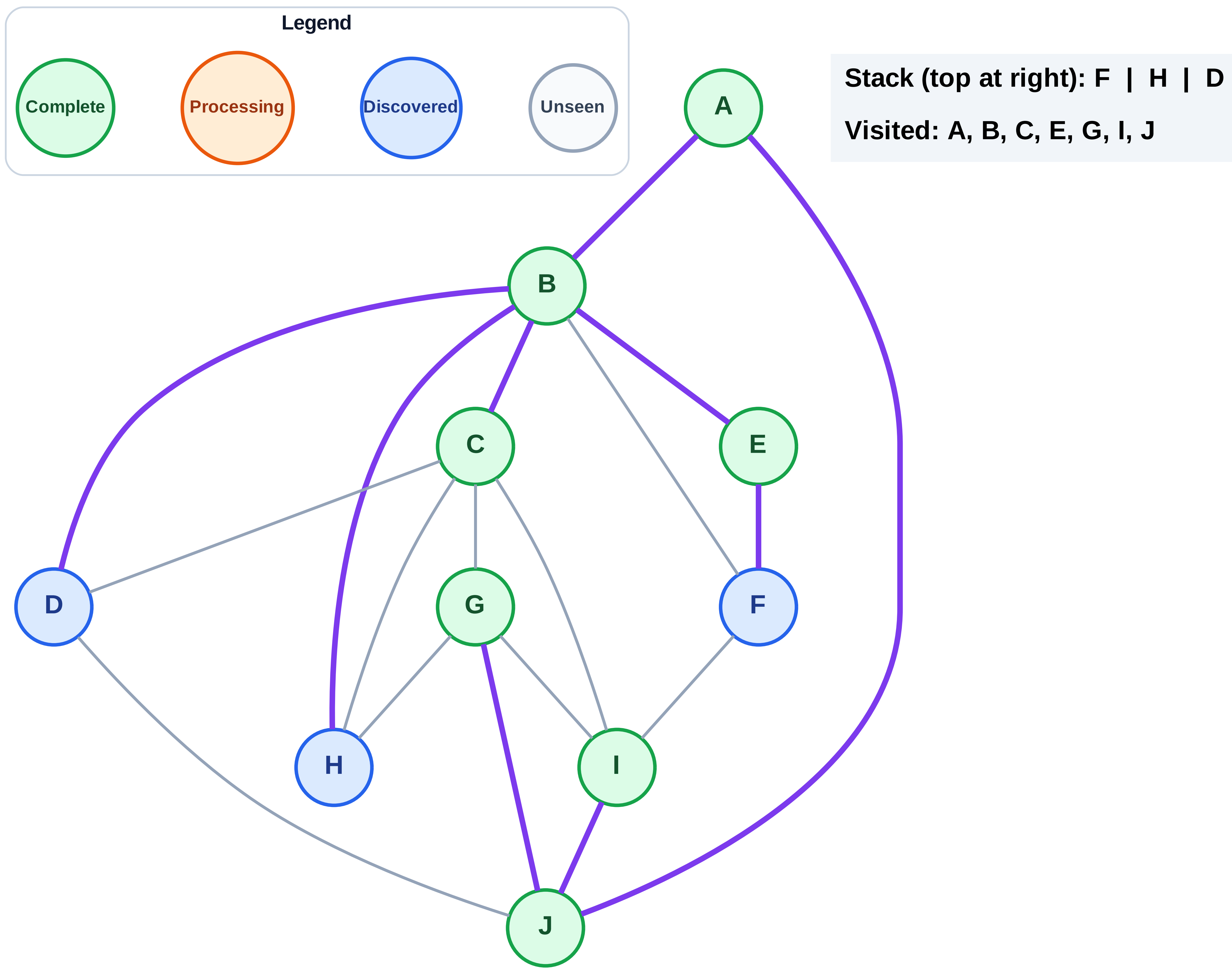
Visited: A, B, E, G, I, J



DFS Traversal
Step 14: Process node C
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): F | H | D
Visited: A, B, E, G, I, J
A
B
C
D
E
F
G
H
I
J

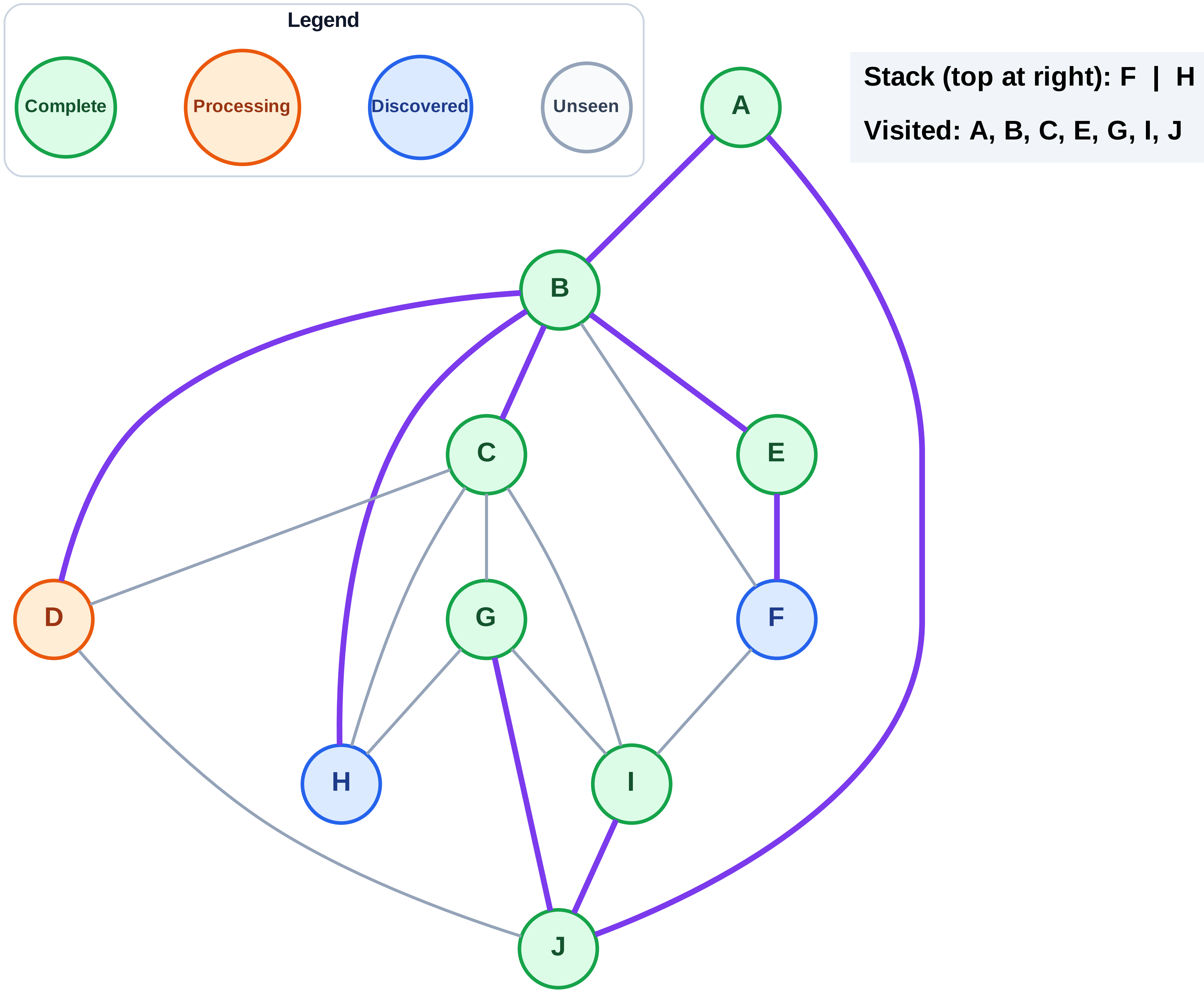
DFS Traversal

Step 15: No new neighbors from C



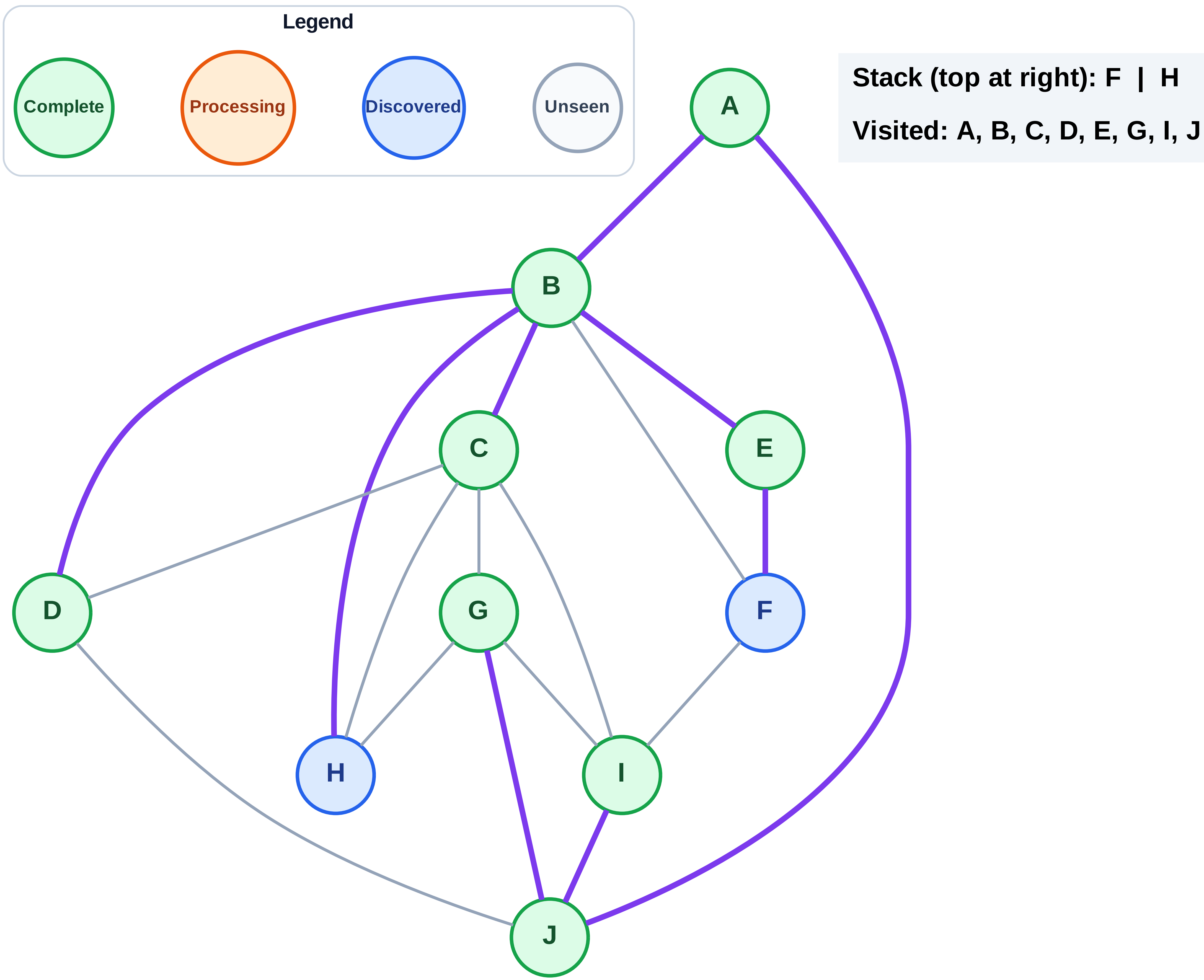
DFS Traversal

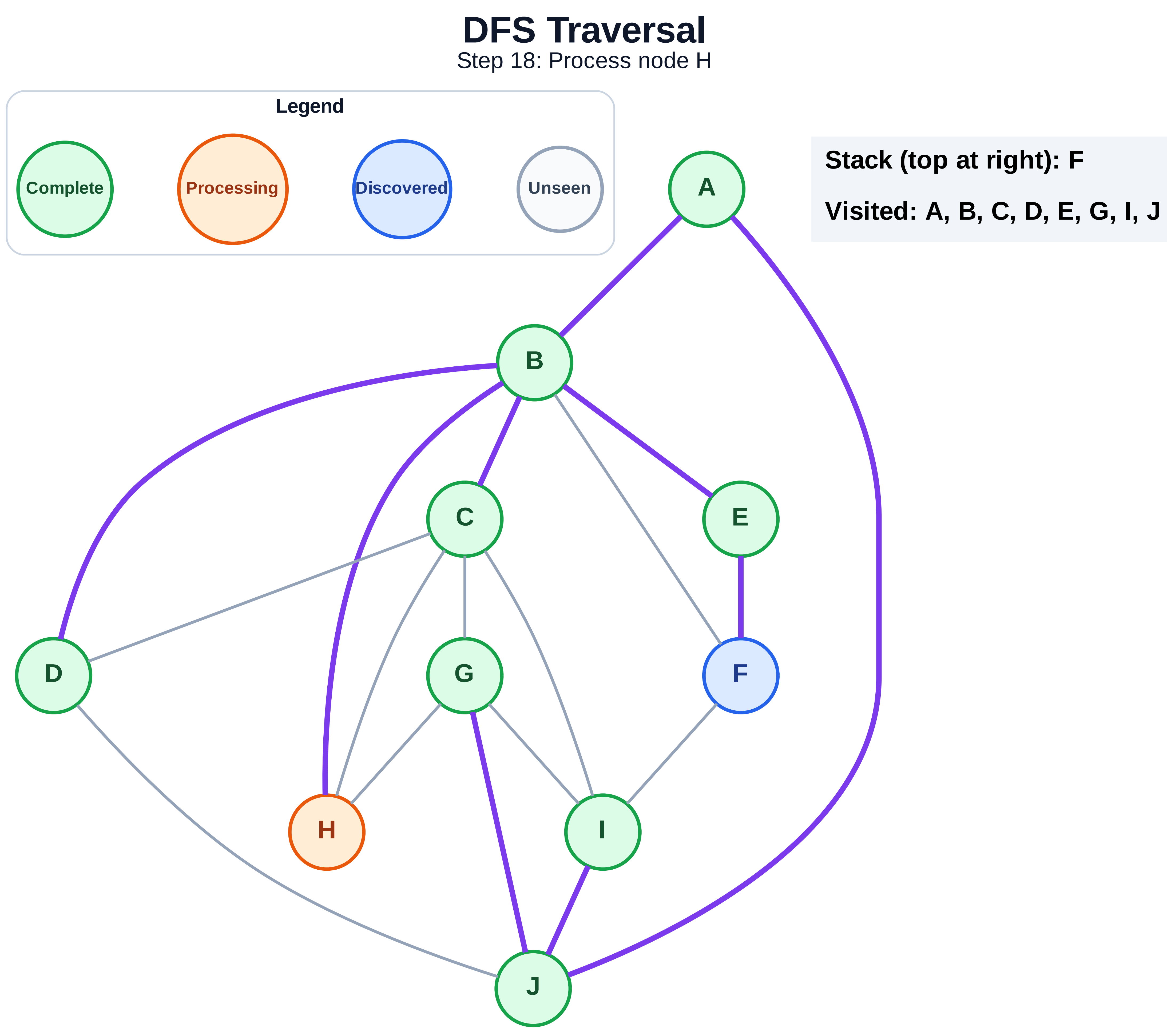
Step 16: Process node D



DFS Traversal

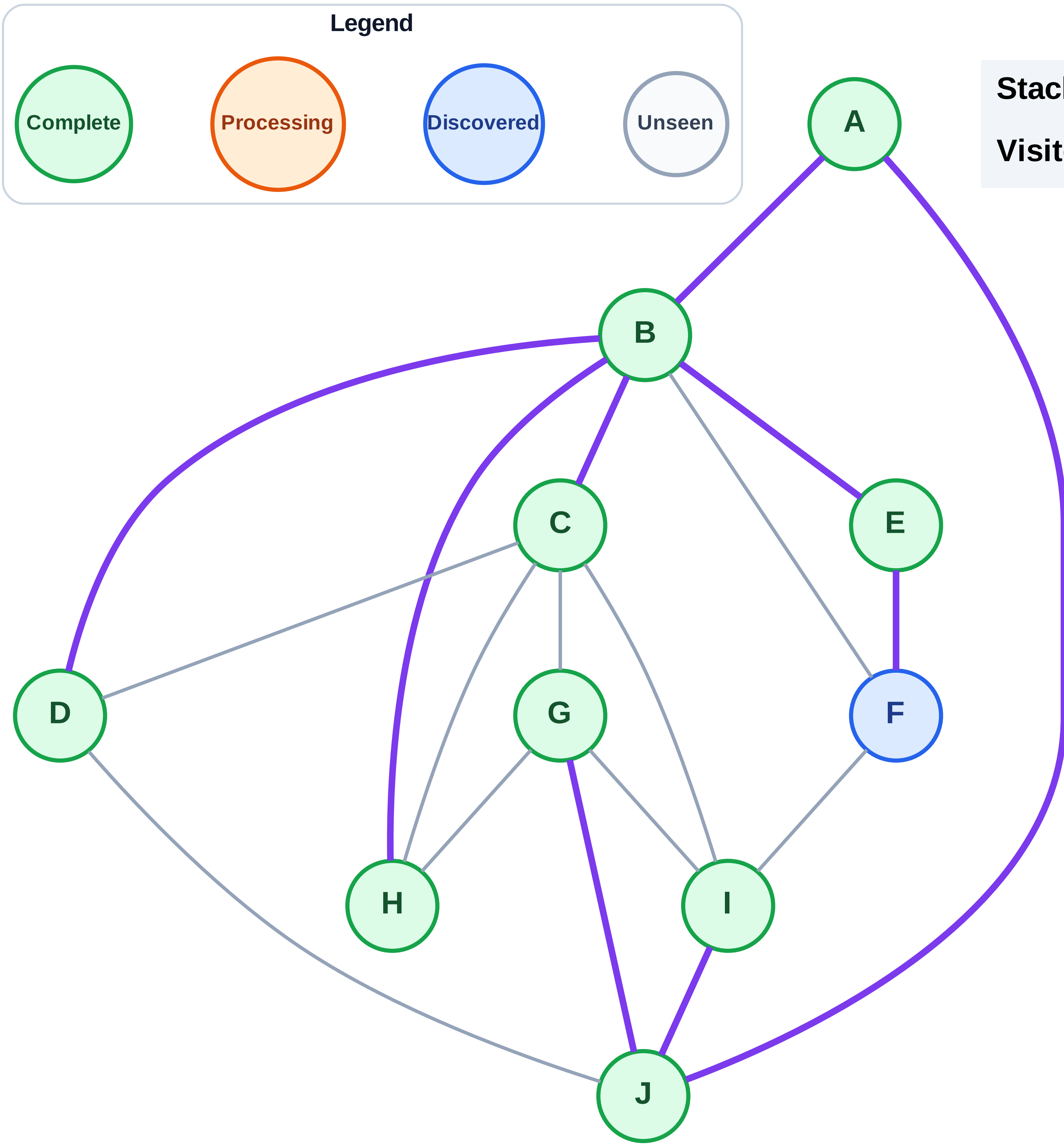
Step 17: No new neighbors from D




DFS Traversal
Step 18: Process node H
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): F
Visited: A, B, C, D, E, G, I, J
A
B
C
D
E
F
G
H
I
J

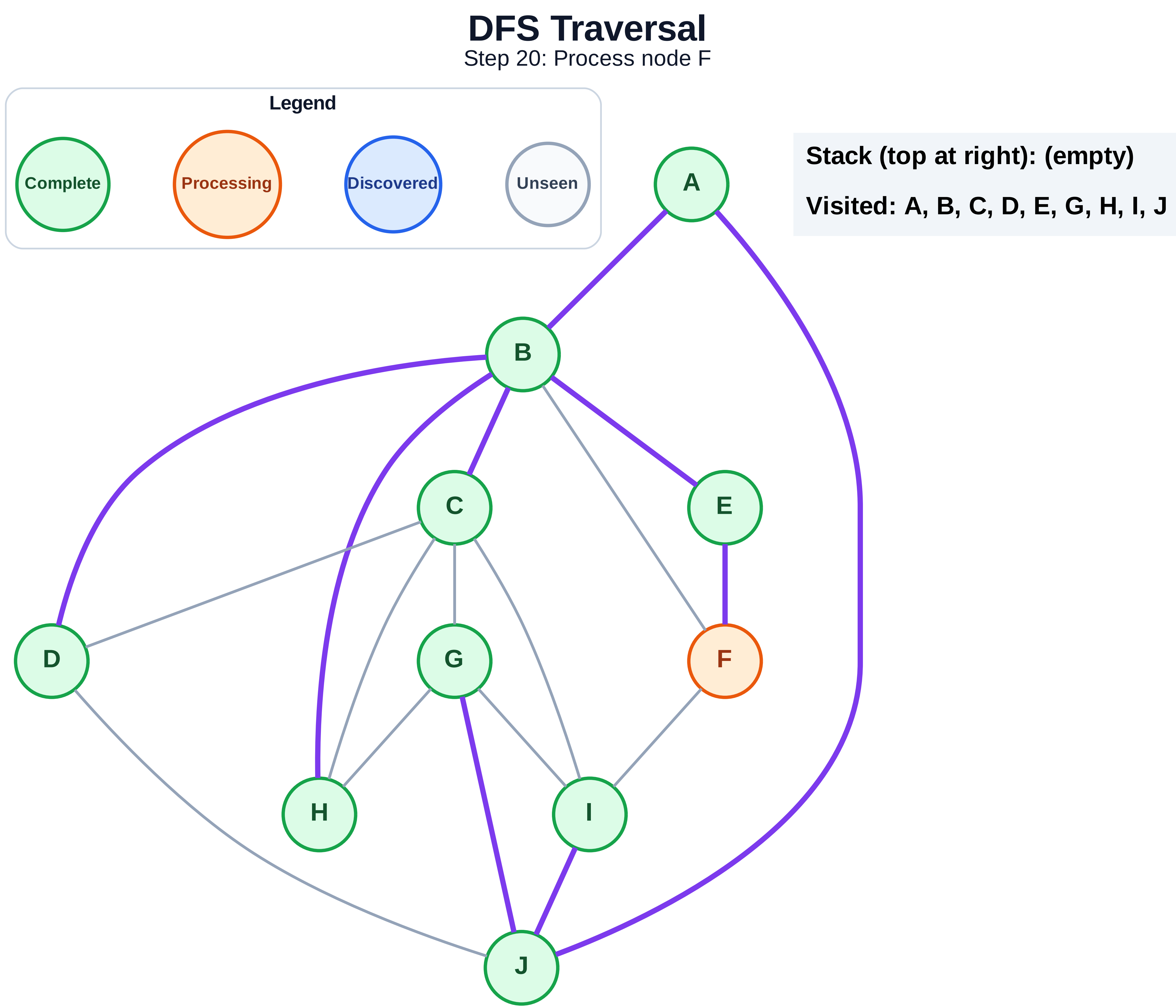
DFS Traversal

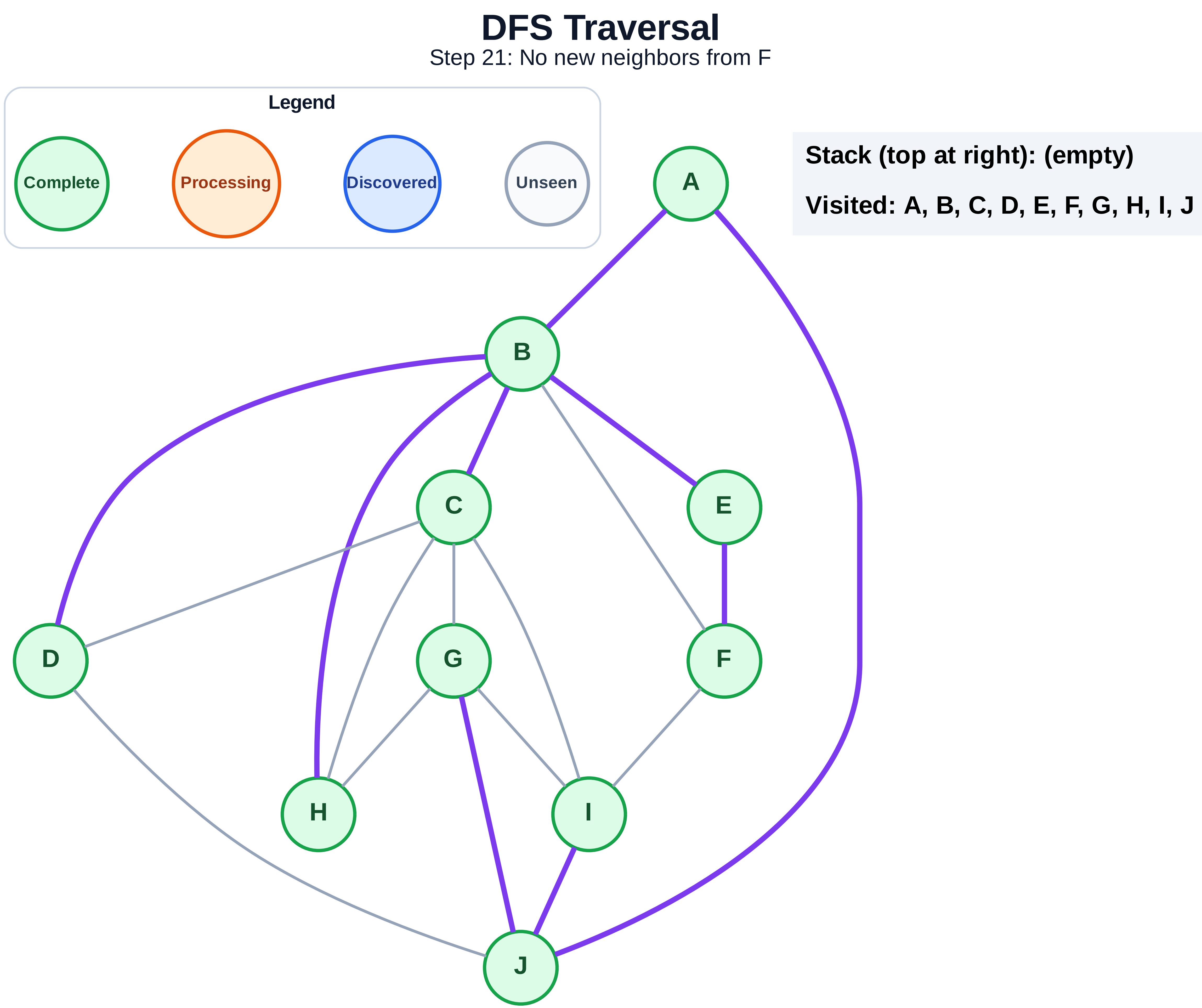
Step 19: No new neighbors from H



Stack (top at right): F

Visited: A, B, C, D, E, G, H, I, J


DFS Traversal
Step 20: Process node F
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: A, B, C, D, E, G, H, I, J
A
B
C
D
E
F
G
H
I
J



DFS Traversal
Step 21: No new neighbors from F
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I, J
A
B
C
E
D
G
F
H
I
J